MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, CMOS, DECODERS, ENCODER, AND DECODERS/DEMULTIPLEXERS MONOLITHIC SILICON, POSITIVE LOGIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, CMOS logic microcircuits. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-Bit latch/4-to-16 line decoder
02	4-Bit latch/4-to-16 line decoder
03	8-Bit priority encoder
04	Dual binary to 1 of 4 decoder/multiplexers
05	Dual binary to 1 of 4 decoder/demultiplexers

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8", dual-in-line)
F	F-5 (16-lead, 1/4" x 3/8", flat-pack)
J	D-3 (24-lead, 1/2" x 1-1/4", dual-in-line)
K	F-6 (24-lead, 3/8" x 5/8" flat-pack)
X	See figure 1 (24-lead, 1/2" x 1-1/4", dual-in-line)
Y	See figure 1 (24-lead, 5/8" x 3/4", flat-pack)
Z	F-5 (16-lead, 1/4" x 3/8", flat-pack except A dimension = 0.1" (2.54 mm) maximum)
U	F-6 (24-lead, 3/8" x 5/8", flat-pack, except A dimension = 0.1" (2.54 mm) maximum)

NOTES:
1. As an exception to 3.5.6.2.3 of MIL-M-38510, for case outlines Z and U only, the leads of bottom brazed ceramic packages (i.e., configuration 2 of case outlines F-5 or F-6) may have electroless nickel undercoating which shall be 50 to 200 microinches (1.27 to 5.08 µm) thick provided the lead finish is hot solder dip (i.e., finish letter A) and provided that, after any lead forming, an additional hot solder dip coating is applied which shall extend from the outer tip of the lead to no more than 0.015 inch (0.38 mm) from the package edge.
2. For bottom or side braded packages, case outlines Z and U only, the S_1 dimension may go to .000 inch (.00 mm) mimimum.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: National Aeronautics and Space Administration, George C, Marshall Space Flight Center, ATTN: EG02, Marshall Space Flight Center, Alabama 35812 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +18 V dc
Input current (each input)	±10 mA
Input voltage range	$(V_{SS} -0.5) \leq V_I \leq (V_{DD} +0.5)$
Storage temperature range	-65°C to +175°C
Maximum power dissipation (P_D) 1/	200 mW
Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case:	(see MIL-M-38510 appendix C)
Junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage (V_{DD} - V_{SS})	4.5 V dc minimum to 5.5 V dc maximum
Input low (V_{IL}) voltage range	0-1.5 V dc @ V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}, 0-2.0 V dc @ V_{DD} = 10 V dc, 0-4.0 V dc, @ V_{DD} = 15 V dc,
Input high (V_{IH}) voltage range	3.5-5.0 V dc @ V_{DD} = 5 V dc, V_{OL} = 10% V_{DD}, V_{OH} = 90% V_{DD}, 8.0-10.0 V dc @ V_{DD} = 10 V dc, 11.0-15 V dc @ V_{DD} = 15 V dc
Ambient operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein. Although eutectic die bonding is preferred, epoxy die bonding may be performed. However, the resin used shall be Dupont 5504 Conductive Silver Paste, or equivalent which is cured at 200°C ±10°C for a minimum of 2 hours. The use of equivalent epoxies or cure cycles shall be approved by the qualifying activity. Equivalency shall be demonstrated in data submitted to the qualifying activity for verification.

3.2.1 Terminal connections. The terminal connections shall be as specified on figure 2.

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3.2.2 Logic diagrams. The logic diagrams shall be as specified on figure 3.

3.2.3 Truth tables. The truth tables shall be as specified on figure 4.

3.2.4 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained and available upon request.

3.2.5 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended ambient operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.6.1. Total dose radiation hardness identifier. Total dose radiation hardness identifier shall be in accordance with MIL-M-38510 and 4.5.5 herein.

3.6.2 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.3 Correctness of indexing and markings. All devices shall be subjected to the final electrical tests specified in table II after part number marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 39 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Delete the sequence specified in 3.1.9 through 3.1.13 of method 5004 and substitute lines 1 through 7 of table II herein.

b. Burn-in tests (method 1015 of MIL-STD-883).

(1) Static tests (test condition A) using circuit shown on figure 5 or equivalent. Ambient temperature (T_A) shall be 125°C minimum. Test duration for each static test shall be 24 hours minimum for class S devices and in accordance with table I of method 1015 for class B devices.

(2) Dynamic test (test condition D) using circuit shown on figure 6 or equivalent. Ambient temperture (T_A) shall be 125°C minimum. Test duration shall be in accordance with table I of method 1015.

c. Interim and final electrical parameters shall be as specified in table II herein.

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1 Propagation delay

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ 125°C Unless otherwise specified	Device type	Limits		Units
				Min	Max	
Positive clamping input to V_{DD}	$V_{IC(pos)}$	T_A = 25°C, V_{DD} = GND, V_{SS} = Open, Output = Open, I_{IN} = 1 mA	All	-	1.5	Vdc
Negative clamping input to V_{SS}	$V_{IC(neg)}$	T_A = 25°C, V_{DD} = Open, V_{SS} = GND, Output = Open, I_{IN} = -1 mA	All	-	-6	Vdc
Quiescent supply current	I_{SS} or I_{DD}	V_{DD} = 18 Vdc, V_{IN} = V_{SS} or V_{DD} All input combinations	All	-	-2.5	µA
High level output voltage	V_{OH1}	V_{DD} = 15 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	14.95	-	Vdc
Low level output voltage	V_{OL1}	V_{DD} = 15 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	-	0.05	Vdc
Input high voltage	V_{IH1}	V_{DD} = 5 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	3.5	-	Vdc
	V_{IH2}	V_{DD} = 10 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	7.0	-	Vdc
	V_{IH3}	V_{DD} = 15 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	11.0	-	Vdc

See note at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ 125°C Unless otherwise specified	Device type	Limits		Units
				Min	Max	
Input low voltage	V_{IL1}	V_{DD} = 5 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	-	1.5	Vdc
	V_{IL2}	V_{DD} = 10 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	-	3.0	Vdc
	V_{IL3}	V_{DD} = 15 Vdc See table III $\|I_O\|$ ≤ 1 µA	All	-	4.0	Vdc
Output high (source) current	I_{OH1}	V_{DD} = 5 Vdc V_{OH} = 4.6 Vdc	All	-0.36	-	mAdc
	I_{OH2}	V_{DD} = 15 Vdc V_{OH} = 13.5 Vdc	All	-2.4	-	mAdc
Output low (sink) current	I_{OL1}	V_{DD} = 5 Vdc V_{OL} = 0.4 Vdc	All	0.36	-	mAdc
	I_{OL2}	V_{DD} = 15 Vdc V_{OL} = 1.5 Vdc	All	2.4	-	mAdc
Input leakage 1/ current	I_{IH}	V_{DD} = 18 Vdc Measure inputs sequentially connect all unused inputs to V_{SS}	All	-	45	nA
Input leakage 1/ current	I_{IL}	V_{DD} = 18 Vdc Measure inputs sequentially connect all unused inputs to V_{DD}	All	-	-45	nA

See note at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ 125°C Unless otherwise specified	Device type	Limits Min	Limits Max	Units
Input capacitance	C_i	V_{DD} = 0 Vdc, f = 1 MHz, T_A = 25°C	All	-	7.5	pF
Input test voltage	V_{ZAP}	C_1 = 100 pF R_z = 1.5 kΩ (see 4.5.3)	All	400	-	V
Minimum setup time	t_s	V_{DD} = 5 Vdc, C_L = 50 pF (See figure 7)	01 02		210	ns
Propagation delay times data to output High to low level Low to high level	t_{PHL1} t_{PLH1}	V_{DD} = 5 Vdc C_L = 50 pF R_L = 200 kΩ (See figure 7)	01 02	49	1360	ns
Propagation delay times inhibit to output High to low level Low to high level	t_{PHL2} t_{PLH2}		01 02	25	700	ns
Propagation delay times E1 to EO High to low level Low to high level	t_{PHL1} t_{PLH1}		03	11	310	ns
Propagation delay times E1 to GS High to low level Low to high level	t_{PHL2} t_{PLH2}		03	11	310	ns
Propagation delay times E1 to Qn outputs High to low level Low to high level	t_{PHL3} t_{PLH3}		03	17	480	ns
Propagation delay times Dn to GS High to low level Low to high level	t_{PHL4} t_{PLH4}		03	17	480	ns
Propagation delay times Dn to Qn outputs High to low level Low to high level	t_{PHL5} t_{PLH5}		03	22	620	ns

See note at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = o V -55°C ≤ T_A ≤ 125°C Unless otherwise specified	Device type	Limits Min	Limits Max	Units
Propagation delay times input to output High to low level Low to high level	t_{PHL1} t_{PLH1}	V_{DD} = 5 Vdc C_L = 50 pF R_L = 200 kΩ (See figure 7)	04	22	620	ns
Propagation delay times E to output High to low level Low to high level	t_{PHL2} t_{PLH2}		04	20	560	ns
Propagation delay times input to outputs High to low level Low to high level	t_{PHL1} t_{PLH1}		05	22	620	ns
Propagation delay times E to outputs High to low level Low to high level	t_{PHL2} t_{PLH2}		05	20	560	ns
Transition times High to low level	t_{THL}	V_{DD} = 5 Vdc, C_L = 50 pF R_L = 200 kΩ (See figure 7)	All	10	280	ns
Low to high level	t_{TLH}		All	10	280	ns
Propagation delay times data to output High to low level Low to high level	t_{PHL1} t_{PLH1}	V_{DD} = 10 Vdc R_L = 200 kΩ C_L = 50 pF (See figure 7)	01 02	19	370	ns
Propagation delay times inhibit to output High to low level Low to high level	t_{PHL2} t_{PLH2}		01 02	11	220	ns
Propagation delay times E1 to E0 High to low level Low to high level	t_{PHL1} t_{PLH1}		03	6	110	ns

See note at end of table.

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from the point of entry of the lead into the body of the the package within one lead width

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions V_{SS} = 0 V -55°C ≤ T_A ≤ 125°C Unless otherwise specified	Device type	Limits Min	Limits Max	Units
Propagation delay times E1 to GS High to low level Low to high level	t_{PHL2} t_{PLH2}	V_{DD} = 10 Vdc C_L = 50 pF R_L = 200 kΩ (See figure 7)	03	6	110	ns
Propagation delay times E1 to Qn outputs High to low level Low to high level	t_{PHL3} t_{PLH3}		03	9	170	ns
Propagation delay times Dn to GS High to low level Low to high level	t_{PHL4} t_{PLH4}		03	9	170	ns
Propagation delay times Dn to Qn outputs High to low level Low to high level	t_{PHL5} t_{PLH5}		03	11	220	ns
Propagation delay times input to output High to low level Low to high level	t_{PHL1} t_{PLH1}		04	10	190	ns
Propagation delay times E to output High to low level Low to high level	t_{PHL2} t_{PLH2}		04	9	170	ns
Propagation delay times input to outputs High to low level Low to high level	t_{PHL1} t_{PLH1}		05	10	190	ns
Propagation delay times E to outputs High to low level Low to high level	t_{PHL2} t_{PLH2}		05	9	170	ns
Transition times High to low level Low to high level	t_{THL} t_{TLH}	V_{DD} = 10 Vdc, C_L = 50 pF R_L = 200 kΩ (See figure 7)	All All	5 5	100 100	ns ns

1/ Input current of one input node.

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TABLE II. Burn-in and electrical test requirements.

Line no.	Applicable tests and MIL-STD-883 test methods	Class S device 1/				Class B device 1/			
		Ref. par.	Table 2/ III sub-groups	Table 3/ IV delta limits	Test circuit figure	Ref. par.	Table 2/ III sub-groups	Table 3/ IV delta limits	Test circuit figure
1	Interim electrical parameters (method 5004)		1				1		
2	Static burn-in I (method 1015)	4.2b 4.5.2			5				
3	Same as line 1		1	Δ					
4	Static burn-in II (method 1015)	4.2b 4.5.2			5	4.2b 4.5.2	4/		5
5	Same as line 1		1*	Δ		4.2d	1*	Δ	
6	Dynamic burn-in (method 1015)	4.2b 4.5.2			6				
7	Same as line 1	4.2d	1*	Δ					
8	Final electrical parameters (method 5004)		1*,2,3 7,9				1*,2,3, 7,9		
9	Group A test requirements (method 5005)	4.4.1	1,2,3,4, 7,8,9, 10,11			4.4.1	1,2,3, 4,7,9, 10,11		
10	Group B V_{ZAP} requirement	4.4.2 4.5.3			8	4.4.2 4.5.3			8
11	Group C end-point electrical parameters (method 5005)					4.4.3	1,2,3	Δ	
13	Group D end-point electrical parameters (method 5005)	4.4.4	1,2,3			4.4.4	1,2,3		

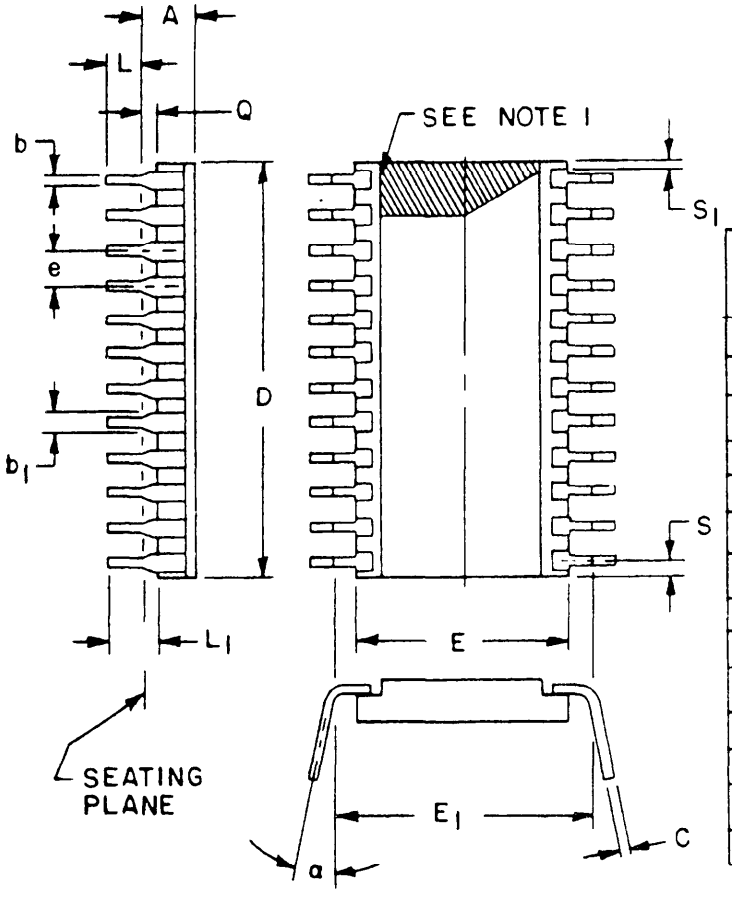
1/ Blank spaces indicate tests are not applicable.

2/ (*) indicates PDA applied to subgroup 1 (see 4.2.1).

3/ (Δ) indicates delta limit shall be required only on table III subgroup 1, where specified, and delta values shall be computed with reference to the previous interim electrical parameters.

4/ The device manufacturer may at his option either perform delta measurements or within 24 hours after burn-in (or removal of bias) perform the final electrical parameter measurements.

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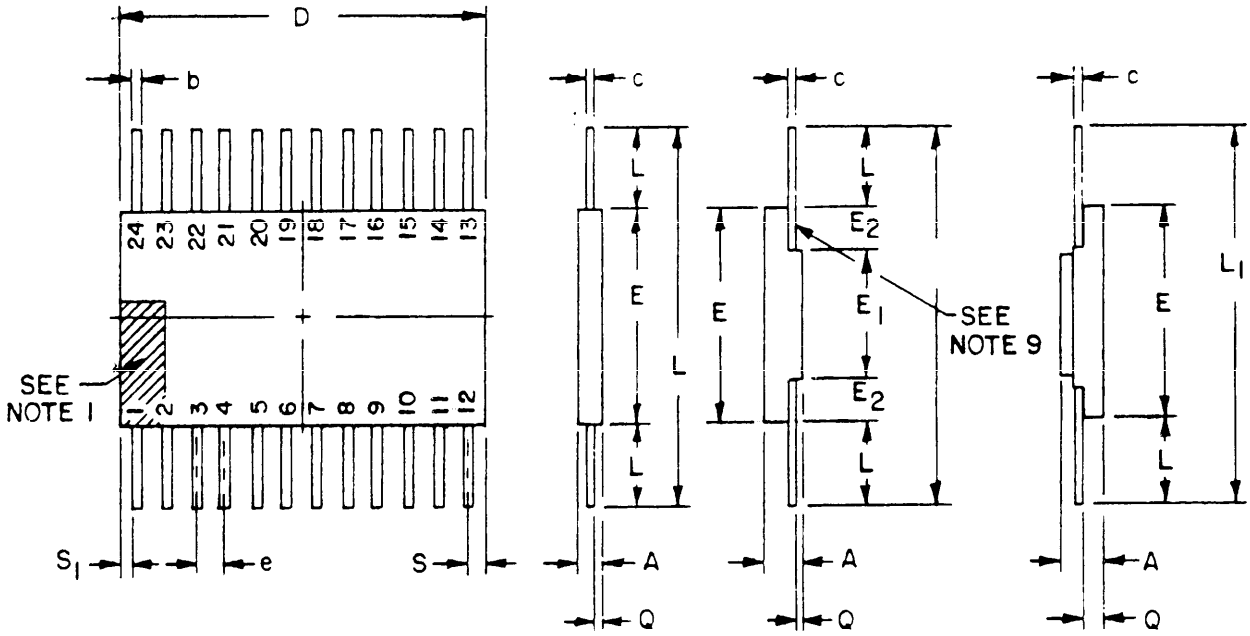


SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A		.225		5.72	
b	.014	.023	.36	.58	8
b_1	.030	.070	.76	1.78	2,8
C	.008	.015	.20	.38	8
D		1.290		32.77	4
E	.450	.610	11.43	15.49	4
E_1	.590	.620	14.99	15.75	7
e	.100 BSC		2.54 BSC		5,9
L	.120	.200	3.05	5.08	
L_1	.150		3.81		
Q	.015	.075	.38	1.90	3
S		.098		2.49	6
S_1	.005		.13		6,10
α	0°	15°	0°	15°	

NOTES:
1. Index area; a notch or a pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. The minimum limit for dimension S_1 may be .020 (.51 mm) for leads number 1, 12, 13, and 24 only.
3. Dimension Q shall be measured from the seating plane to the base plane.
4. This dimension allows for off-center lid, meniscus and glass overrun.
5. The basic pin spacing is .100 (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 (.25 mm) of its exact logitudinal position relative to pins 1 and 24.
6. Applies to all four corners (leads number 1, 12, 13, and 24).
7. Lead center when α is 0°. E_1 shall be measured at the centerline of the leads.
8. All leads.
9. Twenty-two spaces.
10. Dimension S_1 may be .000 (.00 mm) if the leads are brazed to the ceramic body.

FIGURE 1. Case outline X (24-lead, 1/2" x 1-1/4").

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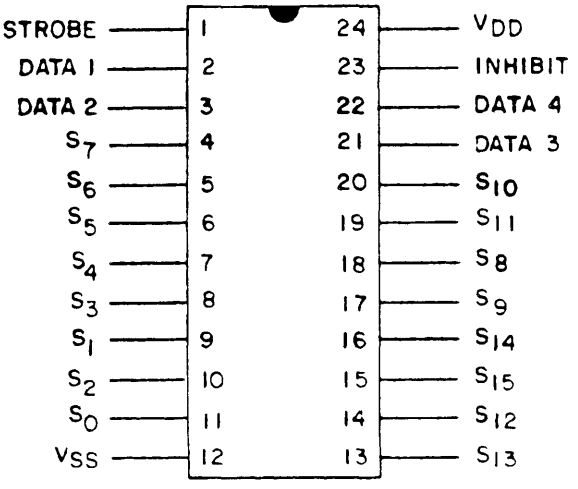
SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	.045	.115	1.14	2.92	
b	.015	.019	.38	.48	5
c	.003	.006	.08	.15	5
D		.725		18.42	3
E	.570	.650	14.48	16.51	3
e	.050 BSC		1.27 BSC		4,6
L	.250	.370	6.35	9.40	
L_1	.860		21.84		
Q	.010	.060	.25	1.52	
S		.045		1.14	7
S_1	.005		.13		7,8

NOTES:
1. Index area; a notch or a pin one identification mark shall be located adjacent to pin one and shall be within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. Dimension Q shall be measured at the point of exit of the lead from the body.
3. This dimension allows for off-center lid, meniscus and glass overrun.
4. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 (.13 mm) of its exact logitudinal position relative to pins 1 and 24.
5. All leads.
6. Twenty-two spaces.
7. Applies to all four corners (leads number 1, 12, 13, and 24).
8. Dimension S_1 may be .000 (.00 mm) if leads number 1, 12, 13, and 24 bend toward the cavity of the package within one lead width from the point of entry of the lead into the body or if the leads are brazed to the metallized ceramic body.
9. Optional configuration. If this configuration is used, no organic or polymeric materials shall be molded to the bottom of the package to cover the leads.

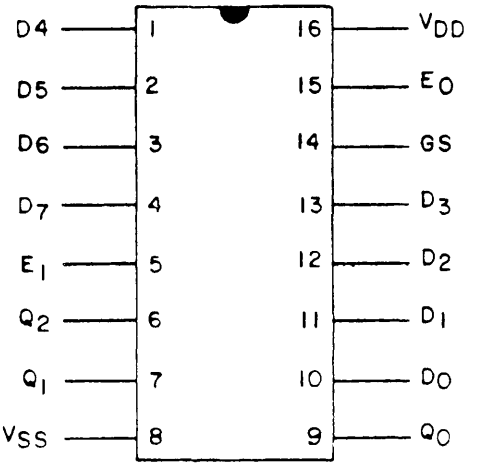
FIGURE 1. Case outline Y (24-lead 5/8" x 3/4") - Continued.

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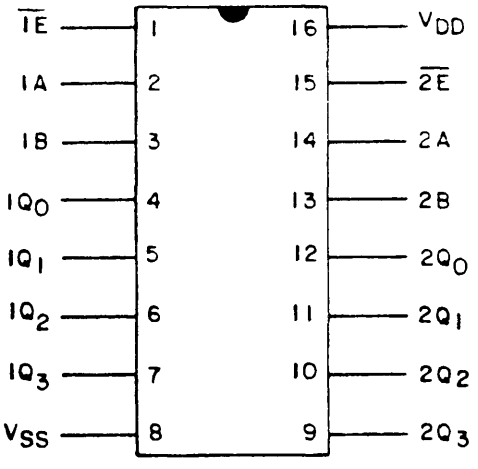
Device types 01 and 02
Cases J, K, X, Y and U



Device type 03
Cases E, F, and Z



Device type 04
Cases E, F, and Z



Device type 05
Cases E, F, and Z

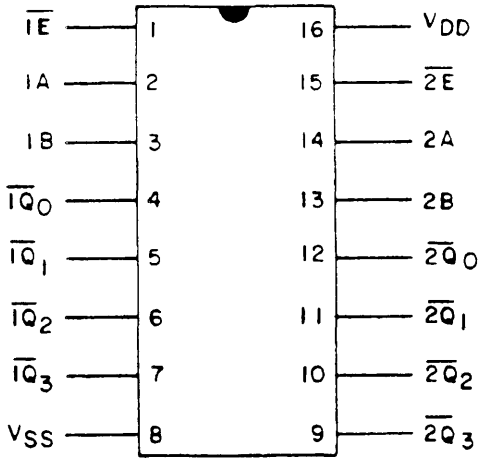


FIGURE 2. Terminal connections.

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DEVICE TYPE 01

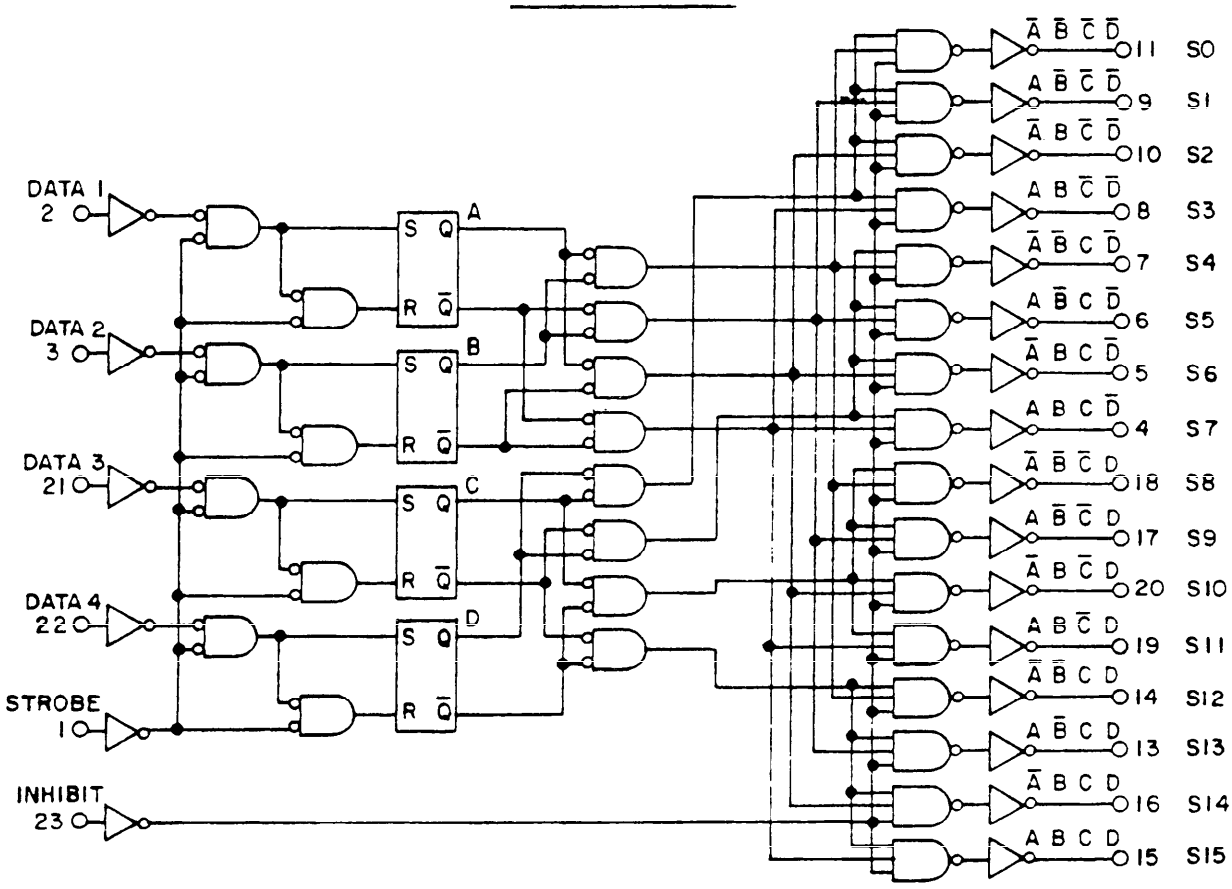


FIGURE 3. Logic diagrams.

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DEVICE TYPE 02

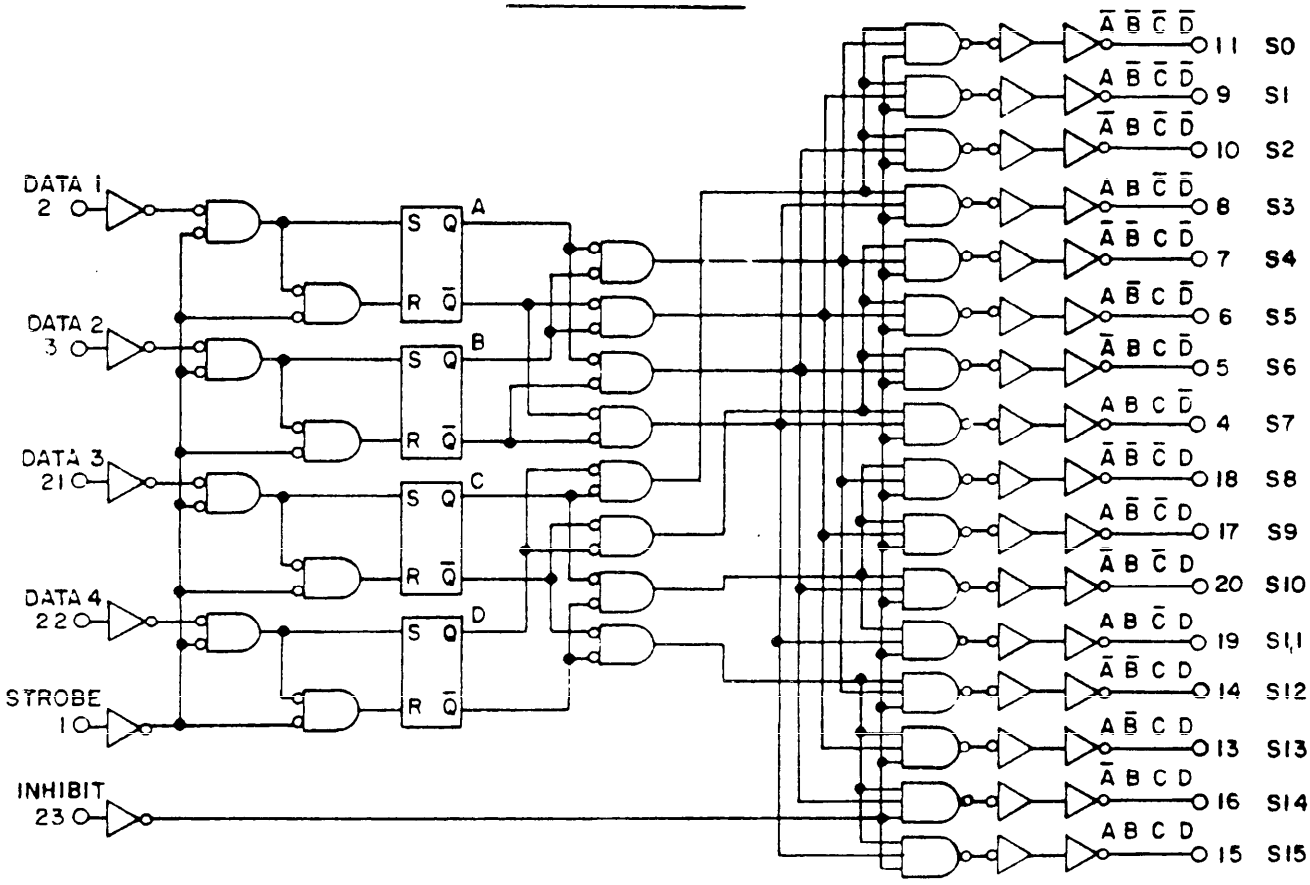


FIGURE 3. Logic diagrams - Continued.

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DEVICE TYPE 03

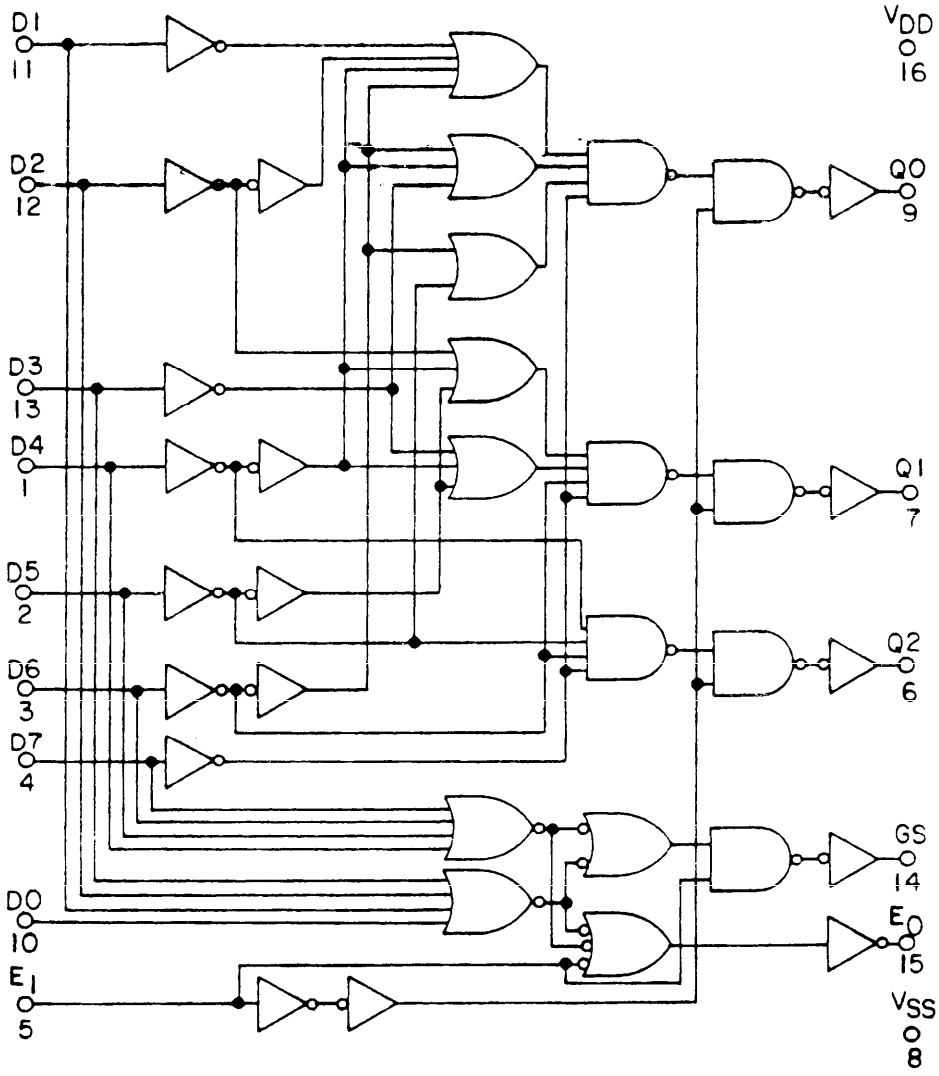
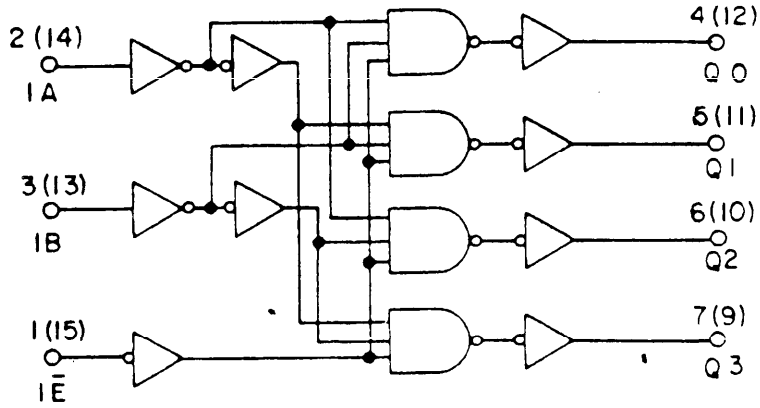


FIGURE 3. Logic diagrams - Continued.

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DEVICE TYPE 04
(ONE OF TWO CIRCUITS SHOWN)



DEVICE TYPE 05
(ONE OF TWO CIRCUITS SHOWN)

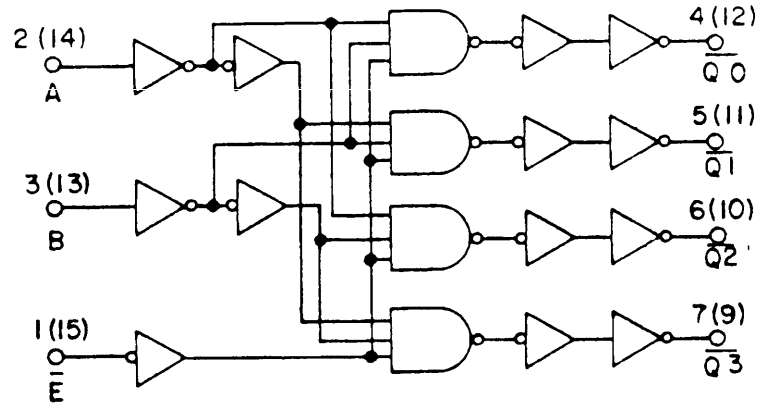


FIGURE 3. Logic diagrams - Continued.

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Device types 01 and 02

DECODE TRUTH TABLE (Strobe = 1)

Inhibit	Decoder inputs				Selected output Type 01 = Logic 1 (High) Type 02 = Logic 0 (Low)
	D_4	D_3	D_2	D_1	
0	0	0	0	0	S0
0	0	0	0	1	S1
0	0	0	1	0	S2
0	0	0	1	1	S3
0	0	1	0	0	S4
0	0	1	0	1	S5
0	0	1	1	0	S6
0	0	1	1	1	S7
0	1	0	0	0	S8
0	1	0	0	1	S9
0	1	0	1	0	S10
0	1	0	1	1	S11
0	1	1	0	0	S12
0	1	1	0	1	S13
0	1	1	1	0	S14
0	1	1	1	1	S15
1	X	X	X	X	All outputs = 0, type 01 All outputs = 1, type 02

X = Don't care Logic 1 = High Logic 0 = Low

Device type 03

Input									Output				
E_1	D7	D6	D5	D4	D3	D2	D1	D0	GS	Q2	Q1	Q0	E_0
0	X	X	X	X	X	X	X	X	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	0	0	1
1	1	X	X	X	X	X	X	X	1	1	1	1	0
1	0	1	X	X	X	X	X	X	1	1	1	0	0
1	0	0	1	X	X	X	X	X	1	1	0	1	0
1	0	0	0	1	X	X	X	X	1	1	0	0	0
1	0	0	0	0	1	X	X	X	1	0	1	1	0
1	0	0	0	0	0	1	X	X	1	0	1	0	0
1	0	0	0	0	0	0	1	X	1	0	0	1	0
1	0	0	0	0	0	0	0	1	1	0	0	0	0

X = Don't Care Logic 1 = High Logic 0 = Low

FIGURE 4. Truth table.

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Device type 04

TRUTH TABLE						
Inputs			Outputs			
E	B	A	Q3	Q2	Q1	Q0
0	0	0	0	0	0	1
0	0	1	0	0	1	0
0	1	0	0	1	0	0
0	1	1	1	0	0	0
1	X	X	0	0	0	0

X = Don't Care
Logic 1 = High; Logic 0 = Low

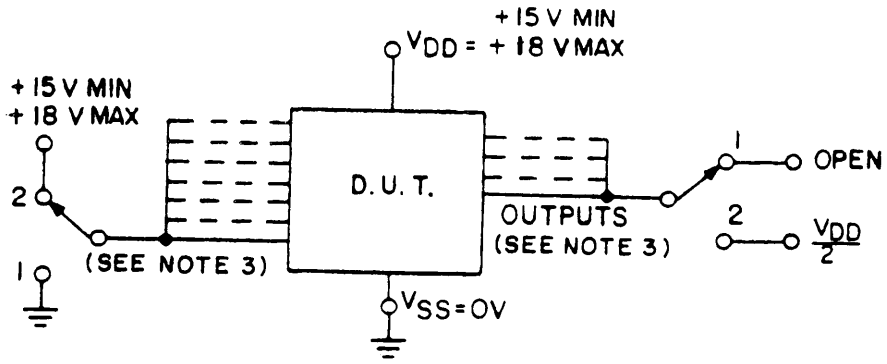
Device type 05

TRUTH TABLE						
Inputs			Outputs			
E	B	A	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	$\overline{Q0}$
0	0	0	1	1	1	0
0	0	1	1	1	0	1
0	1	0	1	0	1	1
0	1	1	0	1	1	1
1	X	X	1	1	1	1

X = Don't Care
Logic 1 = High; Logic 0 = Low

FIGURE 4. Truth table - Continued.

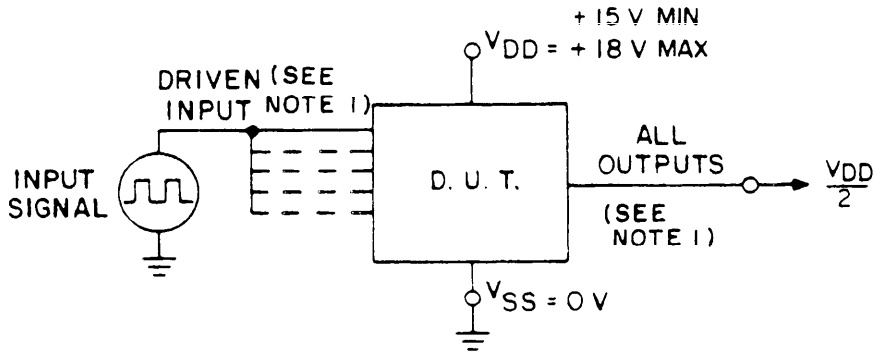
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NOTES:
1. For static burn-in I, all inputs are connected to 0 volts, switch position 1.
2. For static burn-in II, all inputs are connected to V_{DD}, switch position 2.
3. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exeed ±20% of its branded value due to use, heat or age.
4. Output may be in switch position 1 or 2.

FIGURE 5. Static burn-in test circuits.

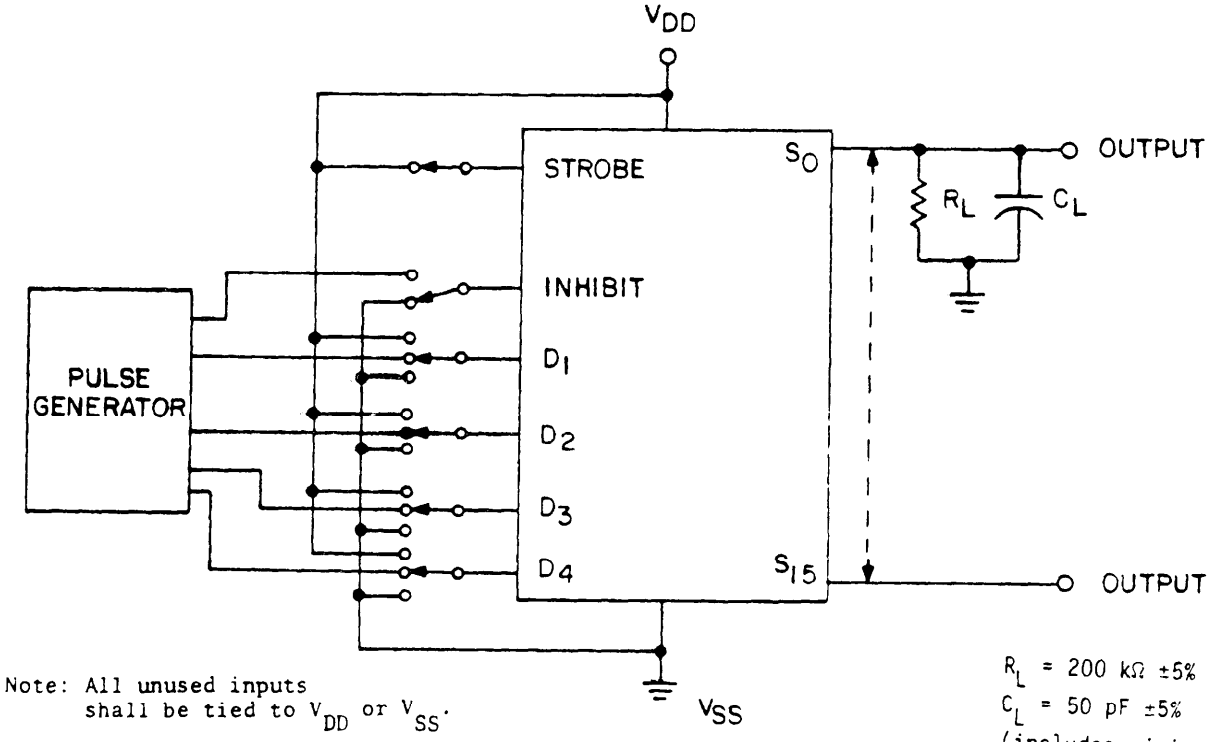
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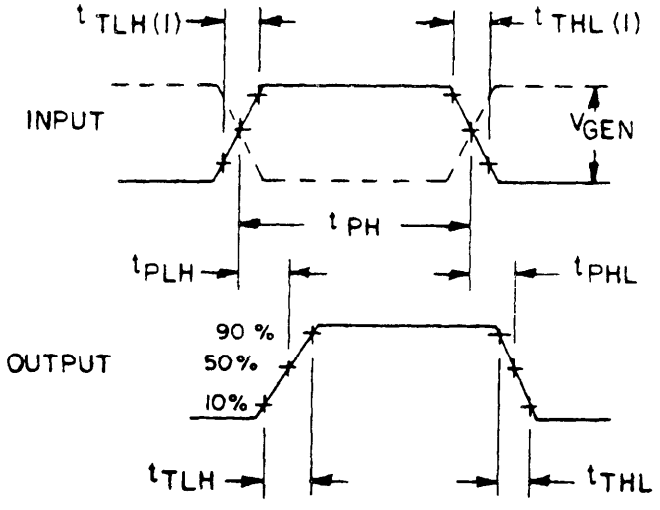


NOTES:
1. Except for V_{DD} and V_{SS}, each terminal shall be connected through a resistor whose value is 2 kΩ to 47 kΩ. The actual measured value of the resistor selected shall not exceed ±20% of its branded value due to use, heat or age.
2. Input signal requirements:
   a. Square wave, 50% duty cycle.
   b. 25 kHz < PRR < 1 MHz.
   c. t_{TLH} and t_{THL} < 1 µs.
   d. Voltage level:
      Minimum = V_{SS} -0.5 V, +10% V_{DD}.
      Maximum = V_{DD} +0.5, -10% V_{DD}.
3. Strobe input shall be connected to V_{DD} and inhibit shall be connected to V_{SS} (for device types 01 and 02 only).
4. Enable input shall be connected to V_{SS} for device types 04 and 05; enable input shall be connected to V_{DD} for device type 03.

FIGURE 6. Dynamic burn-in and steady-state life test circuits.

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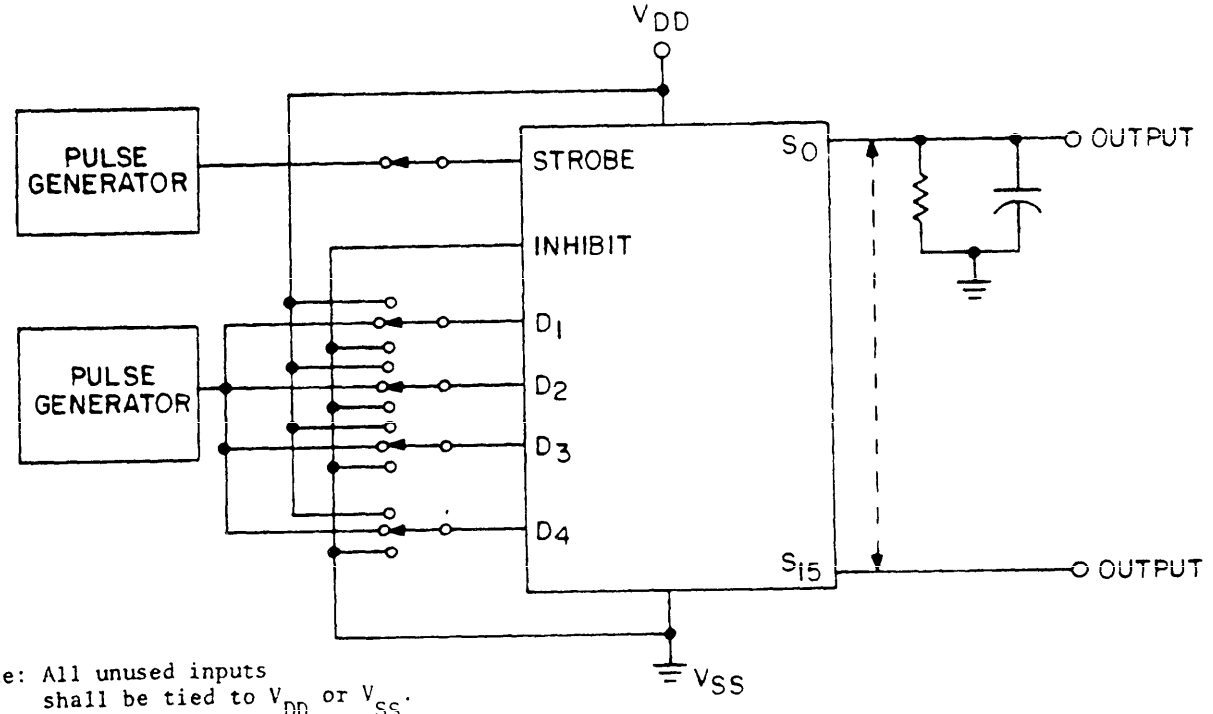




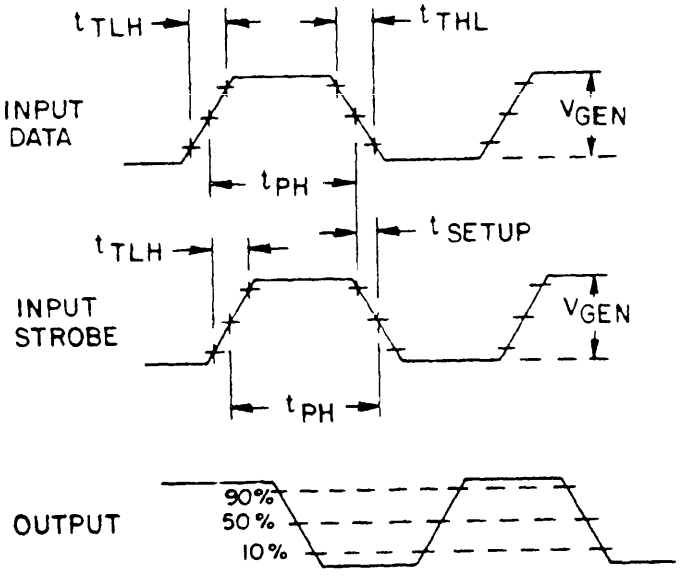
NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD}$ ±1.0%, t_{PH} = 1.0 ±0.1 µs, $t_{THL(1)}$ and $t_{TLH(1)}$ = 20 ±2 ns and PRR = 500 kHz.
2. All outputs shall have the same load circuit as shown for the S_0 output.
3. See table III for complete terminal conditions.

FIGURE 7. Switching time test circuit and waveforms for device types 01 and 02.

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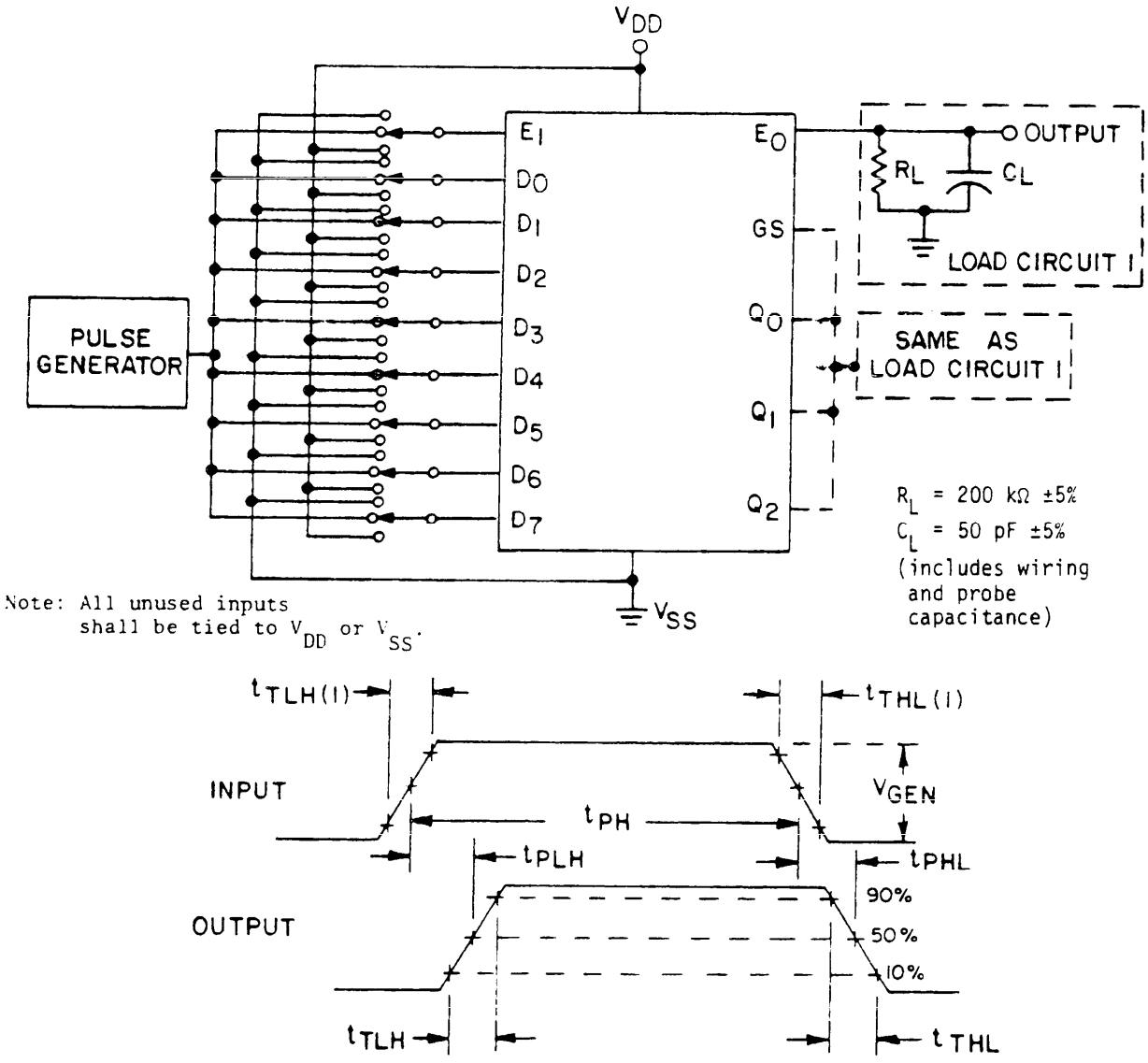
Note: All unused inputs shall be tied to V_{DD} or V_{SS}.



NOTES:
1. The pulse generators have the following characteristics: $V_{GEN} = V_{DD} \pm 1.0\%$, $t_{PH} = 1.0\ \mu s \pm 0.1\ \mu s$, t_{THL} and $t_{TLH} = 20 \pm 2$ ns and PRR = 500 kHz.
2. Requirements for setup times are considered met if proper output state changes occur with t_{SETUP} set to that given in the limits column of table III.

FIGURE 7. Switching time circuit and waveforms for device types 01 and 02 - Continued.

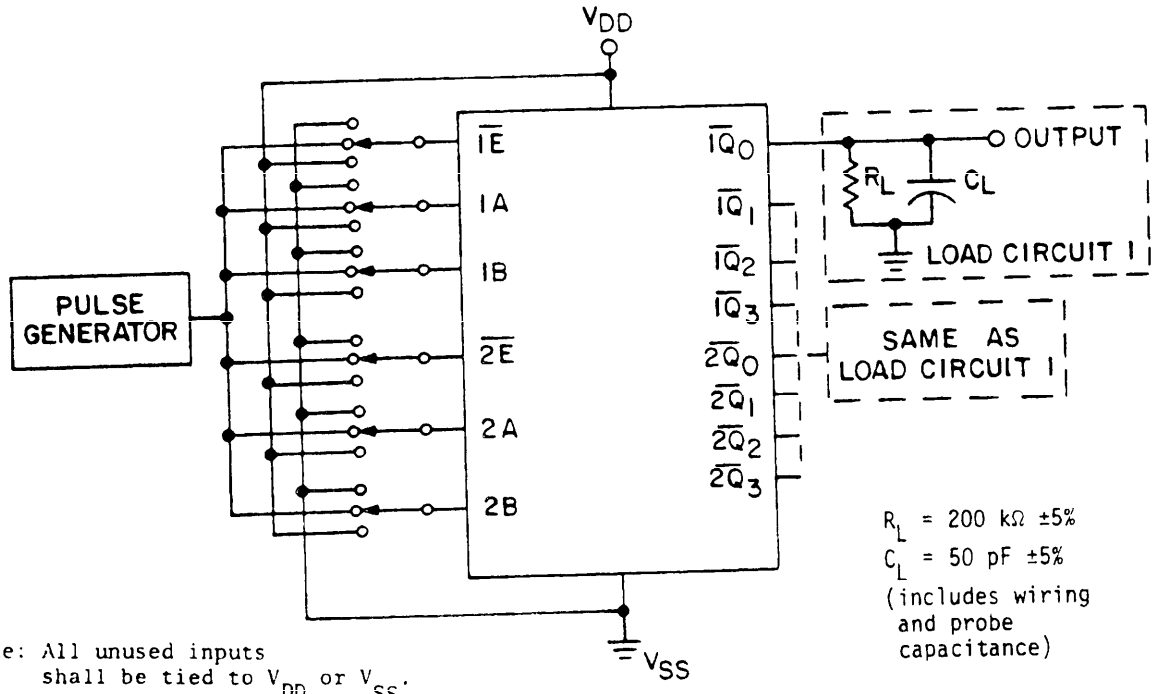
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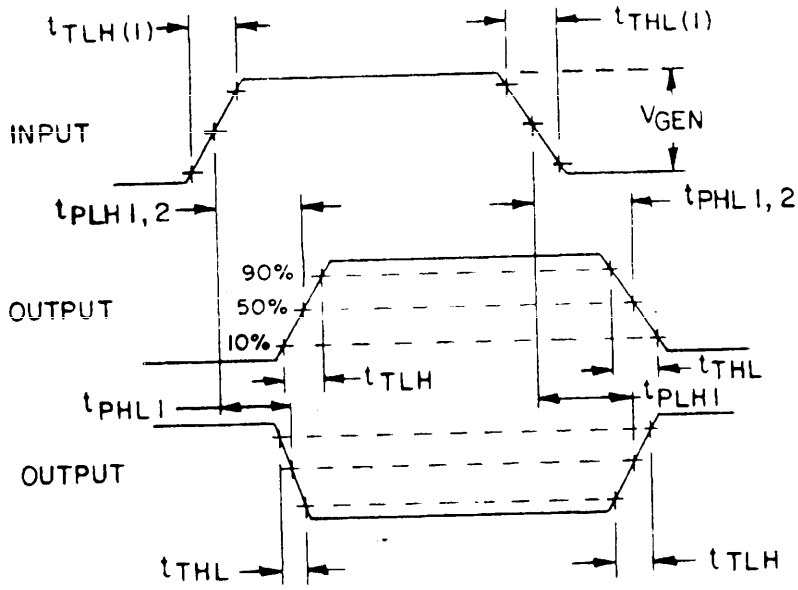
NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, t_{PH} = 1.0 ±0.1 µs, $t_{THL(I)}$ and $t_{TLH(I)}$ = 20 ±2 ns and PRR = 1 MHz.
2. See table III for complete terminal conditions.

FIGURE 7. Switching time test circuit and waveforms for device type 03 - Continued.

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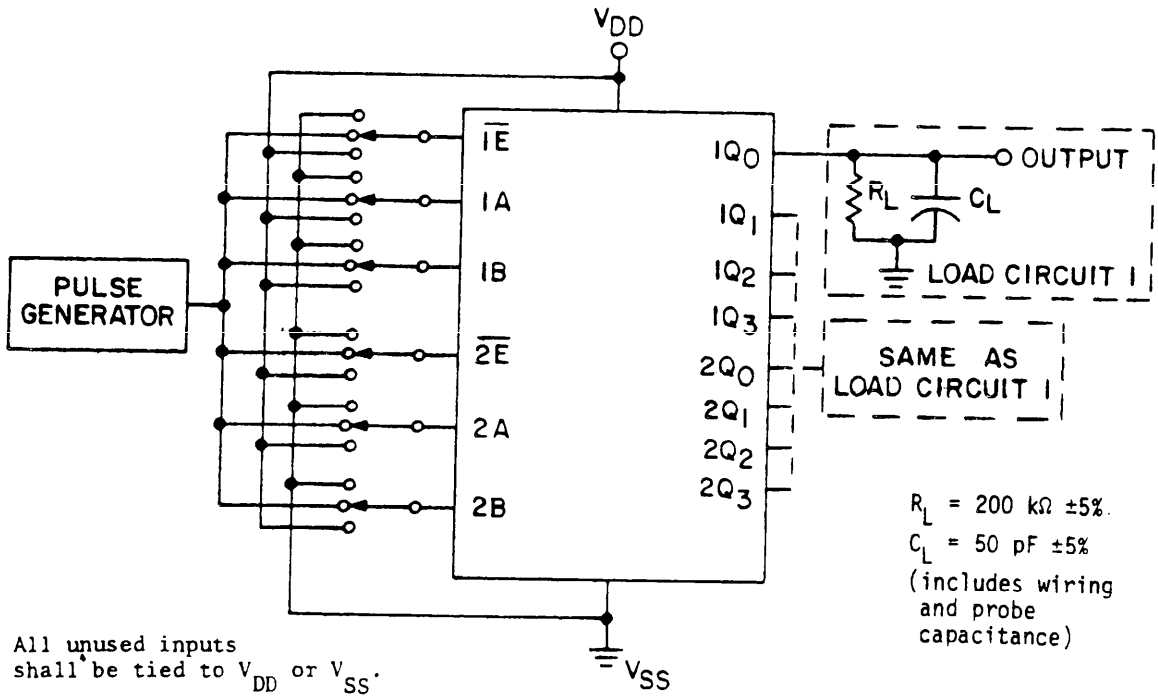
Note: All unused inputs shall be tied to V_{DD} or V_{SS}.



NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD} \pm 1\%$, $t_{PH} = 1.0 \pm 0.1$ µs, $t_{THL(1)}$ and $t_{TLH(1)} = 20 \pm 2$ ns and PRR = 1 MHz.
2. See table III for complete terminal conditions.

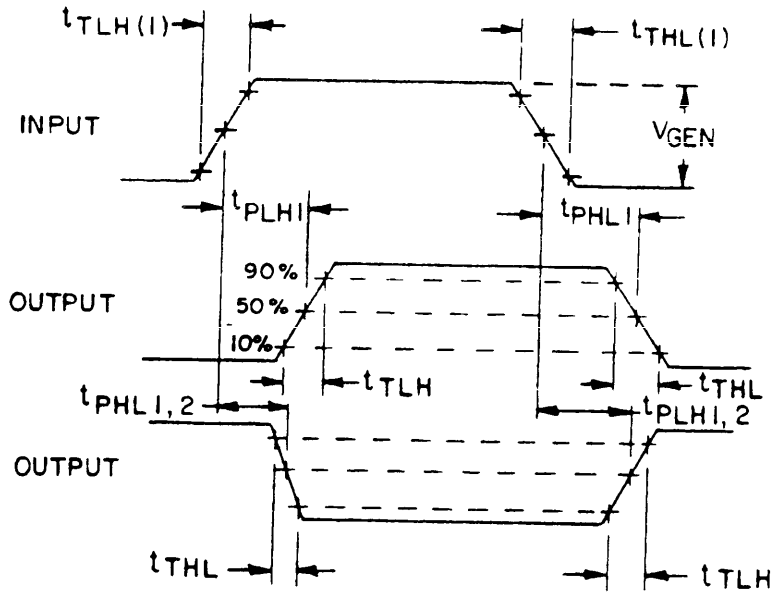
FIGURE 7. Switching time test circuit and waveforms for device type 05 - Continued.

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Note: All unused inputs shall be tied to V_{DD} or V_{SS}.

R_L = 200 kΩ ±5%.
C_L = 50 pF ±5%
(includes wiring and probe capacitance)



NOTES:
1. The pulse generator has the following characteristics: $V_{GEN} = V_{DD}$ ±1%, t_{PH} = 1.0 ±0.1 µs, $t_{THL(1)}$ and $t_{TLH(1)}$ = 20 ±2 ns and PRR = 1 MHz.
2. See table III for complete terminal conditions.

FIGURE 7. Switching time test circuit and waveforms for device type 04 - Continued.

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TABLE III. Group A inspection for device type 01

Symbol	MIL-STD-883 method	Cases J,K X, Y, U	Terminal conditions 1/																							Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	I	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																															
V_{IC} (pos)		1 2 3 4 5 6	1mA	1mA	1mA																		1mA	1mA	1mA	GND	ST D1 D2 D3 D4 I		1.5					Vdc
V_{IC} (neg)		7 8 9 10 11 12	-1mA	-1mA	-1mA									GND									-1mA	-1mA	-1mA		ST D1 D2 D3 D4 I		-6.0					
I_{SS} 2/	3005	13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49	18V GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND GND 18V 18V 18V 18V GND	GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND 18V GND GND 18V 18V 18V 18V GND	GND GND 18V 18V GND GND 18V 18V GND GND 18V 18V GND GND 18V 18V GND GND 18V 18V GND GND 18V 18V GND GND 18V 18V GND GND 18V 18V GND GND									GND										GND 18V 18V 18V 18V GND 18V 18V 18V 18V GND 18V 18V 18V 18V GND 18V 18V 18V 18V GND	GND 18V 18V GND GND 18V 18V GND GND 18V 18V GND GND	GND 18V GND	18V	V_{SS}		-.75		-2.5			uA
V_{OH1}	3006	50 51 52 53 54 55	15V GND 15V GND 15V GND	GND 15V GND 15V GND 15V	GND GND 15V 15V GND GND																		GND 15V	GND		15V	S0 S1 S2 S3 S4 S5	14.95		14.95		14.95		Vdc

See notes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued

Symbol	MIL-STD-883 method	Cases J,K, ℓ, Y, U	Terminal conditions 1/																							Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	I	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
V_{OH1}	3006	56	15V	GND	15V									GND									15V	GND	GND	15V	S6	14.95		14.95		14.95		Vdc
		57		15V	15V																		15V	GND			S7							
		58		GND	GND																		GND	15V			S8							
		59		15V	GND																						S9							
		60		GND	15V																						S10							
		61		15V	15V																						S11							
		62		GND	GND																						S12							
		63		15V	GND																						S13							
		64		GND	15V																						S14							
		65		15V	15V																		↓	↓	↓	↓	S15							
V_{OL1}	3007	66	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	↓	15V	↓	15V	↓	S0		0.05		0.05		0.05	
		67																									S1							
		68																									S2							
		69																									S3							
		70																									S4							
		71																									S5							
		72																									S6							
		73																									S7							
		74																									S8							
		75																									S9							
		76																									S10							
		77																									S11							
		78																									S12							
		79																									S13							
		80																									S14							
		81																									S15							
V_{IL1}		82	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/		3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	All Outputs	3/	3/	3/	3/	3/	3/	
V_{IL2}		83																																
V_{IL3}		84																																
V_{IH1}		85																																
V_{IH2}		86																																
V_{IH3}		87																																
I_{OL1}		88	GND	GND	GND	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V		0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	0.4V	GND	GND	5.0V	5.0V	S7	0.51		0.36		0.64		mA
		89																									S6							
		90																									S5							
		91																									S4							
		92																									S3							
		93																									S1							
		94																									S2							
		95																									S0							
		96																									S13							
		97																									S12							
		98																									S15							
		99																									S14							
		100																									S9							
		101																									S8							
		102																									S11							
		103																									S10							

See notes at end of table

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases J,K / Z,Y,U	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Subgp 1 T_A = 25°C Min	Max	Subgp 2 T_A = 125°C Min	Max	Subgp 3 T_A = -55°C Min	Max	Units
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{DD}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	1	V_{DD}								
		Test No.																																
I_{OL2}		104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119	GND	GND	GND	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	GND	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	GND	GND	15V	15V	S7 S6 S5 S4 S3 S1 S2 S0 S13 S12 S15 S14 S9 S8 S11 S10	3.4		2.4		4.2		mA
I_{OH1}		120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135	5V	GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V	GND GND 5V 5V GND GND 5V 5V GND GND 5V 5V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V		4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	5V 5V 5V 5V GND 5V	5V	GND	5V	S0 S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15	-0.51		-0.36		-0.64		
I_{OH2}		136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151	15V	GND 15V GND 15V GND 15V GND 15V GND 15V GND 15V GND 15V	GND GND 15V 15V GND GND 15V 15V GND GND 15V 15V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V		13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	GND 15V GND 15V	GND 15V	GND	15V	S0 S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15	-3.4		-2.4		-4.2		
I_{IH1} 4/	3010	152	18V	18V	18V																		18V	18V	18V	18V	All inputs together		6					nA

See notes at end of table.

MIL-M-38510/121B

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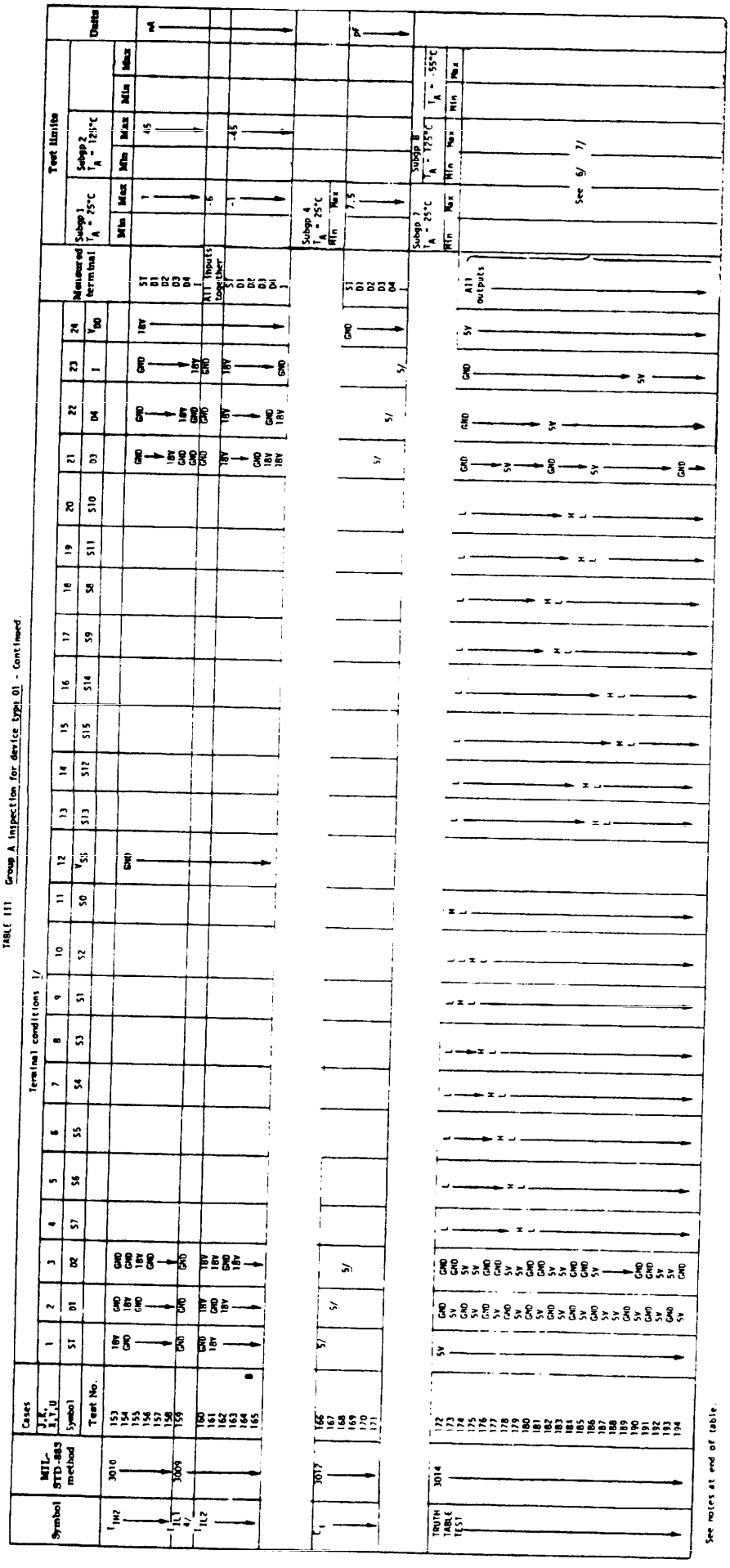


TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,L,U	Test No.	Terminal conditions 1/																					Measured terminal	Test limits						Units
				1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
																													Min	Max	Min	Max	Min	Max	
I_{IH2}	3010		153	18V	GND	GND	ST	S5	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	GND	18V	ST		1		45			nA
			154	GND	18V	GND	D1								GND													D1						
			155			GND	D2								18V													D2						
			156			GND									GND													D3						
			157	GND	GND										18V													D4						
			158												GND													All inputs together						
I_{IL1} 4/	3009		159	GND	GND	GND	GND								18V													S1		-6		-45		
I_{IL2}			160	18V	18V	18V									GND													D1						
			161	GND	GND	GND									18V													D2						
			162	18V	18V	18V									GND													D3						
			163												18V													D4						
			164												GND													All outputs						
			165												18V																			
I_I	3007		166	5V	5V	5V									GND													S1		7.5				mA
			167																									D1						
			168																									D2						
			169																									D3						
			170																									D4						
			171																															
TRUTH TABLE TEST	3014		172	5V	GND	GND	L	L	L	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V	All outputs	Subgp 7 T_A = 25°C		Subgp 8 T_A = 125°C		Subgp 9 T_A = -55°C		
			173		5V	GND	H	L	L	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V		Min	Max	Min	Max	Min	Max	
			174		GND	5V	L	H	L	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			175		5V	5V	H	H	L	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			176		GND	GND	L	L	H	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			177		5V	GND	H	L	H	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			178		GND	5V	L	H	H	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			179		5V	5V	H	H	H	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			180		GND	GND	L	L	L	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			181		5V	GND	H	L	L	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			182		GND	5V	L	H	L	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			183		5V	5V	H	H	L	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			184		GND	GND	L	L	H	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			185		5V	GND	H	L	H	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			186		GND	5V	L	H	H	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			187		5V	5V	H	H	H	H	L	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			188		GND	GND	L	L	L	L	H	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			189		5V	GND	H	L	L	L	H	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			190		GND	5V	L	H	L	L	H	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			191		5V	5V	H	H	L	L	H	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			192		GND	GND	L	L	H	L	H	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			193		5V	GND	H	L	H	L	H	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								
			194		GND	5V	L	H	H	L	H	L	L	L	GND	L	L	L	L	L	L	L	L	L	L	GND	5V								

See 6/ 7/

See notes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases J, K, X, Y, U	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Subgp 7 T_A = 25°C Min	Subgp 7 T_A = 25°C Max	Subgp 8 T_A = 125°C Min	Subgp 8 T_A = 125°C Max	Subgp 8 T_A = -55°C Min	Subgp 8 T_A = -55°C Max	Units
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	1	V_{DD}								
		Test No.																																
TRUTH TABLE TEST	3014	195	5V	GND	GND	L	L	L	L	L	L	L	L	GND	L	L	L	L	L	L	L	L	GND 5V	5V GND	5V	5V	All outputs	See 6/ 7/						
		196		5V	5V																													
		197		GND	5V																													
		198		5V	GND																													
		199		GND	GND																													
		200		5V	5V																													
		201		GND	5V																													
		202		GND	GND																													
		203		GND	5V																													
		204	GND														H						GND 5V											
		205	GND		5V								H											5V										
		206	5V										L										GND		GND									
		207	GND																															
		208	GND	GND	GND																													
																												Subgp 9 T_A = 25°C Min	Subgp 9 T_A = 25°C Max	Subgp 10 T_A = 125°C Min	Subgp 10 T_A = 125°C Max	Subgp 11 T_A = -55°C Min	Subgp 11 T_A = -55°C Max	
t_{SETUP} 8/	Fig 7	209	IN	IN	GND								OUT										GND IN	GND IN	GND	5V	B1 to S0		150		210		150	ns
		210		IN	IN																		GND				B0 to S1							
		211		GND	IN																		IN				B2 to S2							
		212		IN	5V																		IN				B1 to S3							
		213		GND	GND																		IN				B0 to S4							
		214		5V	GND																		IN				B2 to S5							
		215		GND	5V																		IN				B1 to S6							
		216		5V	5V	OUT																	IN				B0 to S7							
		217		GND	GND		OUT																5V				B2 to S8							
		218		5V	5V			OUT															5V				B1 to S9							
		219		GND	5V				OUT											OUT								B0 to S10							
		220		IN	GND					OUT											OUT							B2 to S11							
		221		GND	GND						OUT											OUT						B1 to S12							
		222		5V	GND							OUT																B0 to S13							
		223		GND	IN								OUT															B2 to S14							
		224	↓	5V	5V																							B1 to S15							
t_{PHL1} & t_{PLH1}	3003 Fig 7	225	5V	IN	GND								OUT										GND	GND			B1 to S0	49	970	68	1360	49	970	ns
		226		IN	GND																		IN				B0 to S1							
		227		GND	IN																		IN				B2 to S2							
		228		IN	5V																		IN				B1 to S3							
		229		GND	GND																		IN				B0 to S4							
		230		5V	GND																		IN				B2 to S5							
		231		GND	5V																		IN				B1 to S6							
		232		5V	5V	OUT																	IN				B0 to S7							
		233		GND	GND		OUT																GND				B2 to S8							
		234		5V	GND			OUT															IN				B1 to S9							
		235		GND	5V				OUT											OUT								B0 to S10							
		236		IN	GND					OUT											OUT							B2 to S11							
		237		GND	5V						OUT											OUT						B1 to S12							
		238		5V	GND							OUT																B0 to S13							
		239		GND	IN								OUT															B2 to S14							
		240		5V	5V																							B1 to S15							

See notes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases J, X / K, Y, U	Terminal conditions 1/																							Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgp 9 T_A = 25°C		Subgp 10 T_A = 125°C		Subgp 11 T_A = -55°		
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	I	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
t_{PHL2} & t_{PLH2}	3003 Fig 7	241	5V	GND	GND								OUT	GND									GND	GND	IN	5V	I to S0	25	500	35	700	25	500	ns
		242		5V	GND																		5V				I to S1							
		243		GND	5V																		GND				I to S2							
		244		5V	5V																		5V				I to S3							
		245		GND	GND																						I to S4							
		246		5V	GND																						I to S5							
		247		GND	5V																						I to S6							
		248		5V	5V	OUT	OUT	OUT	OUT	OUT	OUT	OUT							OUT	OUT	OUT	OUT		GND			I to S7							
		249		GND	GND																			5V			I to S8							
		250		5V	GND																						I to S9							
		251		GND	5V																						I to S10							
		252		5V	5V																						I to S11							
		253		GND	GND																						I to S12							
		254		5V	GND																						I to S13							
		255		GND	5V																						I to S14							
		256		5V	5V																						I to S15							
t_{THL} & t_{TLH}	3004 Fig 7	257		IN	GND								OUT										GND	GND	GND		S0	10	200	14	280	10	200	
		258		IN	GND																		IN				S1							
		259		GND	IN																						S2							
		260		IN	5V																						S3							
		261		GND	GND																						S4							
		262		5V	GND																						S5							
		263		GND	5V																						S6							
		264		5V	5V	OUT	OUT	OUT	OUT	OUT	OUT	OUT							OUT	OUT	OUT	OUT					S7							
		265		GND	GND																		GND				S8							
		266		5V	GND																						S9							
		267		GND	5V																						S10							
		268		IN	5V																		IN				S11							
		269		GND	GND																		IN				S12							
		270		5V	GND																		5V				S13							
		271		GND	IN																		5V				S14							
		272		5V	5V																			IN			S15	10	200	14	280	10	200	
																												Subgp 12 T_A = 25°C						
																												Min	Max					
t_{PLH1} & t_{PHL1}	3003 Fig 7	273	10.0	IN	GND								OUT	GND									GND	GND	GND	10.0V	D1 to S0	19	370					
		274		IN	GND																						D1 to S1							
		275		GND	IN																						D2 to S2							
		276		IN	10.0																						D1 to S3							
		277		GND	GND																						D3 to S4							
		278		10.0	GND																						D3 to S5							
		279		GND	10.0																						D3 to S6							
		280		10.0	10.0	OUT	OUT	OUT	OUT	OUT	OUT	OUT							OUT	OUT	OUT	OUT					D3 to S7							
		281		GND	GND																		GND				D4 to S8							
		282		10.0	GND																						D4 to S9							
		283		GND	10.0																						D4 to S10							
		284		IN	10.0																						D1 to S11							
		285		GND	GND																						D3 to S12							
		286		10.0	GND																						D3 to S13							
		287		GND	IN																						D2 to S14							
		288		10.0	10.0																						D4 to S15							

See notes at end of table.

MIL-M-38510/13B

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TABLE III. Group A inspection for device type 01 - Continued.

Symbol	MIL-STD-883 method	Cases J,K, X,Y,U	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Subgp 12 T_A = 25°C Min	Max	Min	Max	Min	Max	Units
		Symbol	ST	O1	O2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	O3	O4	I	V_{DD}							
		Test No.																																
t_{PHL2} & t_{PLH2}	3003 Fig 7	289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304	10.0	10.0 10.0 GND 10.0 GND GND 10.0 GND 10.0 GND GND GND 10.0 GND 10.0	GND GND 10.0 10.0 GND GND 10.0 10.0 GND GND 10.0 GND 10.0 10.0	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND 10.0 GND 10.0	GND 10.0	IN	10.0V	I to S0 I to S1 I to S2 I to S3 I to S4 I to S5 I to S6 I to S7 I to S8 I to S9 I to S10 I to S11 I to S12 I to S13 I to S14 I to S15	11	220					ns
t_{THL} & t_{TLH}	3004 Fig 7	305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320	0	IN IN GND IN GND GND GND 10.0 GND GND GND GND IN GND 10.0	GND GND IN 10.0 GND GND 10.0 10.0 GND GND 10.0 10.0 GND IN 10.0	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND IN GND IN IN 10.0 10.0	GND IN IN IN 10.0 IN	GND		S0 S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15	5	100					

See notes at end of table.

MIL-M-38510/1136

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1/ Pins not designated may be High level logic, Low level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The following timing sequence and input/output conditions shall apply. These tests require only a summary of attributes data.

	TIME SLOT	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32*	33	34	35	36
INPUTS	ST	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	1	0	1
	INH	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0
	D1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	1	0	1	0	1	0	1	0	1	0	1	0	1	0	1	0	0	1	1	1	0
	D2	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1	1	1	0	0	1	1	0	0	1	1	0	0	1	1	0	0	0	1	1	1	0
	D3	0	0	0	0	1	1	1	1	0	0	0	0	1	1	1	1	1	1	1	1	0	0	0	0	1	1	1	1	0	0	0	0	0	1	1	1	0
	D4	0	0	0	0	0	0	0	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	1	1	1	0
OUTPUTS	S0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0
	S1	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S2	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S3	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S4	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S5	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S6	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S7	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S8	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S9	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S10	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S11	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S12	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S13	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S14	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
	S15	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1

* TIME SLOT 32 and Outputs S0: Inhibit shall go from 1 to 0 level before the strobe input goes from 1 to 0.

Input/output conditions

Test	V_{DD}	Input levels		Output levels	
		1	0	1	0
	Volts			Min	Max
V_{IH1} V_{IL1}	5.0	3.5V	1.5V	4.5V	0.5V
V_{IH2} V_{IL2}	10.0	7.0V	3.0V	9.0V	1.0V
V_{IH3} V_{IL3}	15.0	11.0V	4.0V	13.5V	1.5V

4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

5/ See 4.4.1c.

6/ The Truth Table Tests shall be performed in sequence.

7/ The Truth Table Tests shall be performed at V_{IH} and $V_{DD} \leq 5$ Vdc and ≥ 18 Vdc. L = V_{SS} + 0.50 V maximum and H = V_{DD} -0.50 V minimum.

8/ t_{SETUP} may be performed concurrently with t_{PLH} and t_{PHL}, using the maximum allowed test t_{SETUP}.

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TABLE III. Group A inspection for device type 02

Symbol	MIL-STD-883 method	Cases J,K,U	Test No.	Terminal conditions 1/ 1 ST	2 C1	3 D2	4 S7	5 S6	6 S5	7 S4	8 S3	9 S1	10 S2	11 S0	12 V_{SS}	13 S13	14 S12	15 S15	16 S14	17 S9	18 S8	19 S11	20 S10	21 D3	22 D4	23 f	24 V_{DD}	Measured terminal	Test limits Subgp 1 T_A = 25°C Min	Subgp 1 T_A = 25°C Max	Subgp 2 T_A = 125°C Min	Subgp 2 T_A = 125°C Max	Subgp 3 T_A = -55°C Min	Subgp 3 T_A = -55°C Max	Units
V_{IC} (pos)				1mA	1mA	1mA									GND								1mA	1mA	1mA		ST D1 D2 D3 D4		1.5					Vdc
V_{IC} (neg)				-1mA	-1mA	-1mA									GND								-1mA	-1mA	-1mA		ST D1 D2 D3 D4		-.6					
I_{SS} 2/	3005			18V	GND 18V GND 18V GND 18V

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J,K,X,Y,U	Terminal conditions 1/																							Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	S1	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	1	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																																
V_{OL1}	3007	50	15V	GND	GND									GND									GND	GND	GND	15V	S0		0.05		0.05		0.05	Vdc
		51		15V	15V																		15V				S1							
		52		GND	GND																						S2							
		53		15V	15V																						S3							
		54		GND	GND																		GND	15V			S4							
		55		15V	15V																						S5							
		56		GND	GND																						S6							
		57		15V	15V																						S7							
		58		GND	GND																						S8							
		59		15V	15V																						S9							
		60		GND	GND																						S10							
		61		15V	15V																						S11							
		62		GND	GND																						S12							
		63		15V	15V																						S13							
		64		GND	15V																						S14							
		65		15V	15V																						S15							
V_{OH1}	3006	66																							15V		S0	14.95		14.95		14.95		
		67																									S1							
		68																									S2							
		69																									S3							
		70																									S4							
		71																									S5							
		72																									S6							
		73																									S7							
		74																									S8							
		75																									S9							
		76																									S10							
		77																									S11							
		78																									S12							
		79																									S13							
		80																									S14							
		81																									S15							
V_{TL1} V_{TL2} V_{TL3} V_{TH1} V_{TH2} V_{TH3}		82 83 84 85 86 87	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	3/	All outputs	3/	3/	3/	3/	3/	3/	
I_{OL1}		88	5V	GND	GND								0.4V										GND	GND	GND	5V	S0	0.51		0.36		0.64		mA
		89		5V	GND																		5V				S1							
		90		GND	5V																						S2							
		91		5V	5V																						S3							
		92		GND	GND				0.4V																		S4							
		93		5V	GND					0.4V																	S5							
		94		GND	5V						0.4V																S6							
		95		5V	5V	0.4V	0.4V	0.4V																			S7							
		96		GND	GND							0.4V															S8							
		97		5V	GND								0.4V														S9							
		98		GND	5V														0.4V								S10							
		99		5V	5V															0.4V							S11							
		100		GND	GND																0.4V						S12							
		101		5V	GND																	0.4V					S13							
		102		GND	5V																						S14							
		103		5V	5V																						S15							

See notes at end of device type.

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TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J, K, X, Y, U	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Subgp 1 T_A = 25°C Min	Max	Subgp 2 T_A = 125°C Min	Max	Subgp 3 T_A = -55°C Min	Max	Units
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	I	V_{DD}								
		Test No.																																
I_{OL2}		104 thru 119	15V	GND	GND	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	GND	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	1.5V	GND → 5V → GND → 5V	GND → 5V	GND	15V	S0 S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15	3.4		2.4		4.2		mA
I_{OH1}		120 thru 135	GND	GND	GND	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V		4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	4.6V	GND	GND	5V	5.0V	S7 S6 S5 S4 S3 S1 S2 S0 S13 S12 S15 S14 S9 S8 S11 S10	-0.51		-0.36		-0.64		
I_{OH2}		136 thru 151				13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V		13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V	13.5V			15V	15V	S7 S6 S5 S4 S3 S1 S2 S0 S13 S12 S15 S14 S9 S8 S11 S10	-3.4		-2.4		-4.2		
I_{IH1} 4/	3010	152	18V	18V	18V																		18V	18V	18V	18V	All inputs together		6					nA
I_{IH2}	3010	153 thru 158	18V GND	GND 18V GND	GND GND 18V GND																		GND → 18V GND GND	GND → 18V GND	GND → 18V	18V	ST D1 D2 D3 D4 I		1		45			

See notes at end of device type.

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TABLE III. Group A inspection for device type 02 - Continued

Symbol	MIL-STD-883 method	Cases J, K, X, Y, U	Terminal conditions 1/																						Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	S7	D1	D2*	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	I	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																															
I_{IL1} 4/	3009	159	GND	GND	GND									GND									GND	GND	GND	18V	All inputs together		-6					nA
I_{IL2}	3009	160 161 162 163 164 165	GND 18V	18V GND 18V	18V 18V GND 18V																		18V GND 18V 18V	18V GND 18V	18V GND		ST D1 D2 D3 D4 I		-1		-45			
																												Subgp 4 T_A = 25°C						
																												Min	Max					
C_I	3012	166 167 168 169 170 171	5/	5/	5/									GND									5/	5/	5/	GND	ST D1 D2 D3 D4		7.5					pF
																												Subgp 7 T_A = 25°C		Subgp 8 T_A = 125°C		T_A = -55°C		
																												Min	Max	Min	Max	Min	Max	
TRUTH TABLE TEST	3014	172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203	5V	GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND	GND GND 5V 5V GND GND 5V 5V GND GND 5V 5V GND GND 5V GND GND 5V 5V GND GND 5V 5V GND GND 5V 5V GND GND 5V 5V GND	H L H	H L H	H L H	H L H	H H L H	H L H	H L H	H H L H	L H	GND	H L H	H L H	H L H	H L H	H L H	H L H	H L H	H L H	GND 5V GND 5V GND 5V GND GND	GND 5V GND 5V	GND 5V	5V	All outputs	See 6/ 7/						

See notes at end of device type.

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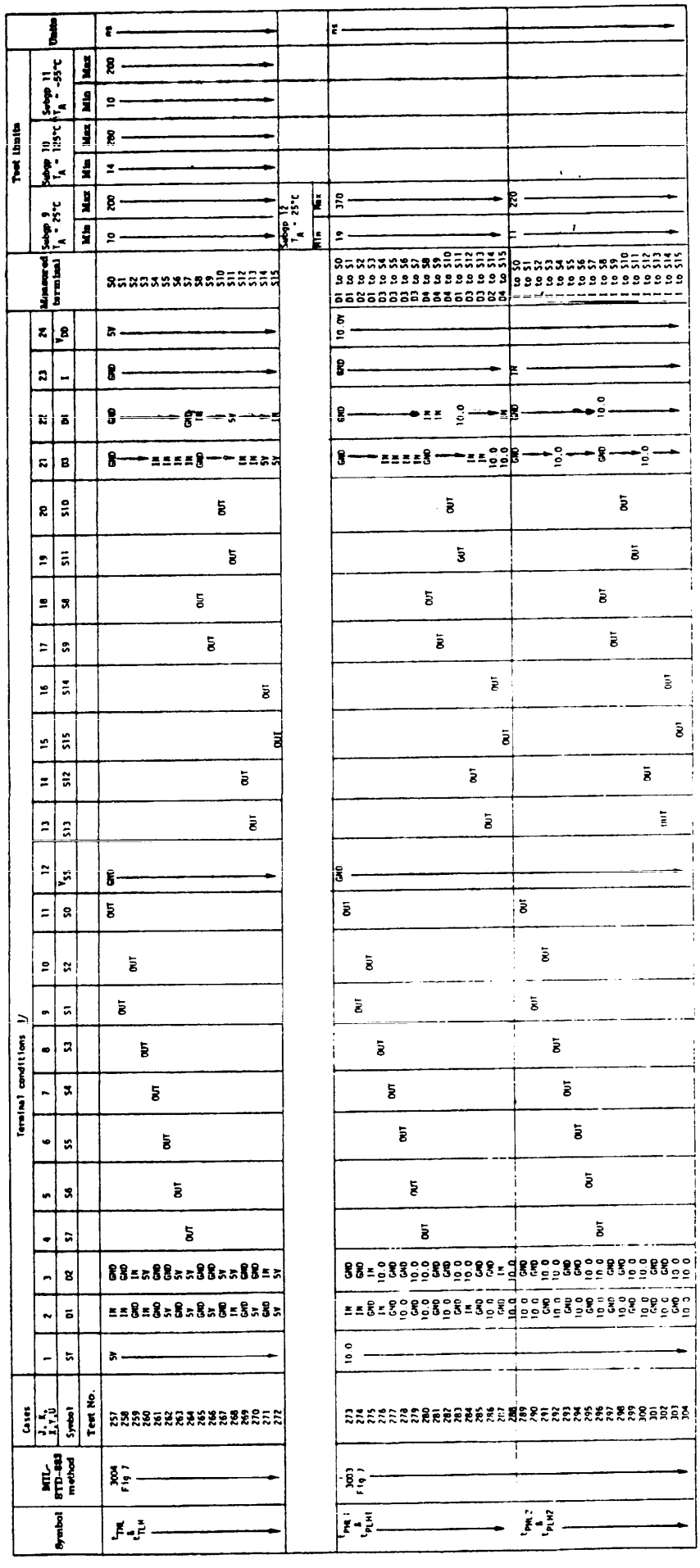
TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J, K, X, 1, U	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test Limits Subgp 7 T_A = 25°C Min	Max	Subgp 8 T_A = 125°C Min	Max	T_A = -55°C Min	Max	Units
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	I	V_{DD}								
		Test No.																																
TRUTH TABLE TEST	3014	204 205 206 207 208	GND GND 5V GND GND	GND 5V 5V 5V GND	GND 5V 5V 5V GND	H	H	H	H	H	H	H	L L H H H	GND	H	H	H ↓ L L	H	H	H	H	H	H	GND 5V 5V 5V GND	GND 5V 5V 5V GND	GND		All outputs		See 6/ 7/					
																												Subgp 9 T_A = 25°C Min	Max	Subgp 10 T_A = 125°C Min	Max	Subgp 11 T_A = -55°C Min	Max	
t_{SETUP} 8/	Fig 7	209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224	IN	IN IN GND IN GND 5V GND 5V GND 5V GND IN GND 5V GND 5V	GND GND IN 5V GND GND 5V 5V GND GND 5V 5V GND GND IN 5V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND ↓ IN ↓ GND ↓ IN IN 5V 5V	GND ↓ IN IN IN 5V ↓ IN	GND	5V	D1 to S0 D1 to S1 D2 to S2 D1 to S3 D3 to S4 D3 to S5 D3 to S6 D3 to S7 D4 to S8 D4 to S9 D4 to S10 D1 to S11 D3 to S12 D3 to S13 D2 to S14 D4 to S15		150		210		150	ns
t_{PHL1} & t_{PLH1}	3009 Fig 7	225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240	5V ↓	IN IN GND IN GND GND 5V 5V GND 5V GND IN GND 5V GND 5V	GND GND IN 5V GND GND 5V 5V GND GND 5V 5V GND GND IN 5V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND ↓ IN ↓ GND ↓ IN IN 5V 5V	GND ↓ IN IN IN 5V ↓ IN	GND	5V	D1 to S0 D1 to S1 D2 to S2 D1 to S3 D3 to S4 D3 to S5 D3 to S6 D3 to S7 D4 to S8 D4 to S9 D4 to S10 D1 to S11 D3 to S12 D3 to S13 D2 to S14 D4 to S15	49	970	68	1360	49	970	
t_{PHL2} & t_{PLH2}	3003 Fig 7	241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256		5V 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V GND 5V	GND GND 5V 5V GND GND 5V 5V GND GND 5V 5V GND GND 5V 5V	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND ↓ 5V ↓ GND ↓ 5V	GND ↓ 5V	IN		I to S0 I to S1 I to S2 I to S3 I to S4 I to S5 I to S6 I to S7 I to S8 I to S9 I to S10 I to S11 I to S12 I to S13 I to S14 I to S15	25	500	35	700	25	500	

See notes at end of device type.

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TABLE III. Group A inspection for device type 02 - Continued.



Symbol	MIL-STD-883 method	Cases J, K, Z, L, U	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Subgp 9 T_A = 25°C Min	Max	Subgp 10 T_A = 125°C Min	Max	Subgp 11 T_A = -55°C Min	Max	Units
		Symbol	S7	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D1	I	V_{DD}								
t_{PHL} & t_{PLH}		Test No.																																
	3004 Fig 7	257	5V	IN	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	OUT	OUT	OUT	OUT	GND	GND	GND	5V	S0	10	200	14	280	10	200	ns
		258		GND	GND																		IN	GND			S1							
		259		IN	IN																		IN	IN			S2							
		260		GND	GND																		IN	IN			S3							
		261		5V	5V																		IN	IN			S4							
		262		GND	GND																		GND	GND			S5							
		263		5V	5V																		IN	IN			S6							
		264		5V	GND																		IN	GND			S7							
		265		GND	GND																		GND	GND			S8							
		266		GND	5V																		IN	10.0			S9							
		267		GND	IN																		IN	GND			S10							
		268		IN	GND																		GND	GND			S11							
		269		GND	GND																		GND	GND			S12							
		270		GND	5V																		IN	10.0			S13							
		271		5V	GND																		IN	GND			S14							
		272		5V	5V																		IN	10.0			S15							

TABLE III. Group A inspection for device type 02 - Continued.

Symbol	MIL-STD-883 method	Cases J, K, Y, Y, U	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Subgp 12 T_A = 25°C Min	Max					Units
		Symbol	ST	D1	D2	S7	S6	S5	S4	S3	S1	S2	S0	V_{SS}	S13	S12	S15	S14	S9	S8	S11	S10	D3	D4	1	V_{DD}								
		Test No.																																
t_{THL} & t_{TLH}	3004 Fig 7	305	10.0	IN	GND								OUT	GND									GND	GND	GND	10.0	S0	5	100					ns
		306		IN	GND																		IN	IN			S1							
		307		GND	IN																		IN	IN			S2							
		308		IN	10.0																		IN	IN			S3							
		309		GND	GND																		GND				S4							
		310		10.0	GND																		↓	↓			S5							
		311		GND	10.0																		IN	IN			S6							
		312		10.0	10.0	OUT																	IN	IN			S7							
		313		GND	GND																		10.0	10.0			S8							
		314		10.0	GND																		↓	↓			S9							
		315		GND	10.0																		IN	IN			S10							
		316		IN	10.0																		10.0	10.0			S11							
		317		GND	GND																		↓	↓			S12							
		318		10.0	GND																		IN	IN			S13							
		319		GND	IN																						S14							
		320		10.0	10.0																						S15							

1/ Pins not designated may be High level logic, Low level logic or open. Exceptions are as follows:
$V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminal shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The following timing sequence and input/output conditions shall apply. These tests require only a summary of attributes data.

	TIME SLOT	0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32* 33 34 35 36
I	ST	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 0 0
N	INH	0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 0
P	D1	0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 1 0 0 1 1 1 0
U	D2	0 0 1 1 0 0 1 1 0 0 1 1 0 0 1 1 1 1 0 0 1 1 0 0 1 1 0 0 1 1 0 0 1 1 1 1 0
T	D3	0 0 0 0 1 1 1 1 0 0 0 0 1 1 1 1 1 1 1 1 0 0 0 0 1 1 1 1 0 0 0 0 1 1 1 1 0
S	D4	0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 1 1 1 1 0 0 0 0 1 1 1 1 0
	S0	0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 1 1 1
	S1	1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
	S2	1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
	S3	1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
	S4	1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
O	S5	1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
U	S6	1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
T	S7	1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
P	S8	1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
U	S9	1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
T	S10	1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
S	S11	1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
	S12	1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
	S13	1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
	S14	1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
	S15	1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0

* Time slot 32 and output S0: Inhibit input shall go from 1 to 0 before the strobe goes from 1 to 0.

Input/output conditions

Test	V_{DD}	Input levels 1	Input levels 0	Output levels 1	Output levels 0
	Volts			Min	Max
V_{IH1} V_{IL1}	5.0	3.5V	1.5V	4.5V	0.5V
V_{IH2} V_{IL2}	10.0	7.0V	3.0V	9.0V	1.0V
V_{IH3} V_{IL3}	15.0	11.0V	4.0V	13.5V	1.5V

4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for each individual input or measure all inputs together.

5/ See 4.4.1c.

6/ The Truth Table Tests shall be performed in sequence.

7/ The Truth Table Tests shall be performed at V_{IH} and $V_{DD} \leq 5$ Vdc and ≥ 18 Vdc. L = V_{SS} +0.50V maximum and H = V_{DD} -0.50V minimum.

8/ t_{SETUP} may be performed concurrently with t_{PLH} or t_{PHL}, using the maximum allowed test t_{SETUP}.

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TABLE III. Group A inspection for device type 03.

Symbol	MIL-STD-883 method	Case E,F,2	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	D4	D5	D6	D7	E1	Q2	Q1	V_{SS}	Q0	D0	D1	D2	D3	GS	E0	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
V_{IC} (pos)		1 2 3 4 5 6 7 8 9	1mA	1mA	1mA	1mA	1mA					1mA	1mA	1mA	1mA			GND	D4 D5 D6 D7 E1 D0 D1 D2 D3		1.5					Vdc
V_{IC} (neg)		10 11 12 13 14 15 16 17 18	-1mA	-1mA	-1mA	-1mA	-1mA			GND		-1mA	-1mA	-1mA	-1mA				D4 D5 D6 D7 E1 D0 D1 D2 D3		-6					
I_{SS} 2/	3005	19 20 21 22 23 24 25 26 27 28	GND 18V GND	GND 18V GND	GND GND GND 18V GND	GND GND 18V GND	GND 18V					GND 18V	GND 18V	GND 18V GND GND	GND 18V GND GND			18V	V_{SS}		-.25		-2.5			uA
V_{OL1}	3007	29 30 31 32 33 34 35 36 37 38	15V GND 15V GND	15V GND 15V GND	15V GND 15V 15V GND	15V GND 15V 15V GND	GND 15V					15V GND 15V	15V GND 15V	15V GND 15V GND GND	15V GND 15V GND			15V	GS Q0 E0 Q0 Q1 Q1 Q2 Q2 Q2 Q2		0.05		0.05		0.05	Vdc
V_{OH1}	3006	39 40 41 42 43 44 45 46 47	GND 15V GND	GND 15V GND	GND 15V 15V GND	GND 15V GND	15V					GND 15V	GND 15V GND	GND 15V GND GND	15V GND				E0 Q0 Q1 Q2 GS GS Q1 Q0 GS	14.95		14.95		14.95		
V_{IL1} V_{IL2} V_{IL3} V_{IH1} V_{IH2} V_{IH3}		48 49 50 51 52 53	3/	3/	3/	3/	3/	3/	3/		3/	3/	3/	3/	3/	3/	3/	3/	All outputs	3/	3/	3/	3/	3/	3/	

See notes at end of device type.

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TABLE III. Group A inspection for Q3 - Continued.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/															Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	D4	D5	D6	D7	E1	Q2	Q1	V_{SS}	Q0	D0	D1	D2	D3	GS	E0	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
I_{OL1}		54 55 56 57 58	GND	GND	GND	GND	GND	0.4V	0.4V	GND	0.4V	GND	GND	GND	GND	0.4V	0.4V	5V	Q2 Q1 Q0 GS E0	0.51		0.36		0.64		mA
I_{OL2}		59 60 61 62 63						1.5V	1.5V		1.5V					1.5V	1.5V	15V	Q2 Q1 Q0 GS E0	3.4		2.4		4.2		
I_{OH1}		64 65 66 67 68				5V GND	5V	4.6V	4.6V		4.6V					4.6V	4.6V	5V	Q2 Q1 Q0 GS E0	-0.51		-0.36		-0.64		
I_{OH2}		69 70 71 72 73				15V GND	15V	13.5V	13.5V		13.5V					13.5V	13.5V	15V	Q2 Q1 Q0 GS E0	-3.4		-2.4		-4.2		
I_{IH1}	3010	74	18V	18V	18V	18V	18V					18V	18V	18V	18V			18V	All inputs together		9					nA
I_{IH2} 4/	3010	75 76 77 78 79 80 81 82 83	18V GND	GND 18V GND	GND GND 18V	GND GND GND 18V GND	GND 18V GND					GND 18V GND GND	GND 18V GND GND	GND 18V GND GND	GND 18V GND				D4 D5 D6 D7 E1 D0 D1 D2 D3		1		45			
I_{IL1}	3009	84					GND					GND	GND	GND	GND				All inputs together		-9					
I_{IL2} 4/		85 86 87 88 89 90 91 92 93	GND 18V	18V GND 18V	18V 18V GND 18V	18V GND 18V	18V GND 18V					18V GND 18V	18V GND 18V	18V GND 18V	18V GND 18V				D4 D5 D6 D7 E1 D0 D1 D2 D3		-1		-45			
																				Subgp 4 T_A = 25°C						
																				Min	Max					
C_1	3012	94 95 96 97 98 99 100 101 102	5/	5/	5/	5/	5/					5/	5/	5/	5/			GND	D4 D5 D6 D7 E1 D0 D1 D2 GS		7.5					pF

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 7 T_A = 25°C		Subgp 8 T_A = 125°C		T_A = -55°C		
		Symbol	D4	D5	D6	D7	E1	C2	Q1	V_{SS}	Q0	D0	D1	D2	D3	GS	E0	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
TRUTH TABLE TEST	3014	103 104 105 106 107 108 109 110 111 112	5V GND 5V ↓ GND	5V GND 5V 5V GND	5V GND 5V 5V GND	5V GND 5V GND	GND 5V	GND GND 5V ↓ GND	GND GND 5V 5V GND GND 5V 5V GND GND	GND	GND GND 5V GND 5V GND 5V GND 5V GND	5V GND 5V	5V GND 5V ↓ GND	5V GND 5V ↓ GND	5V GND 5V ↓ GND GND GND	GND GND 5V	GND 5V GND	5V	All outputs	See 6/ 7/						
																				Subgp 9 T_A = 25°C		Subgp 10 T_A = 125°C		Subgp 11 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL1} & t_{PLH1} t_{PHL2} & t_{PLH2} t_{PHL3} & t_{PLH3}	3003 Fig 7	113, 114, 115 116 117 118 119 120 121 122 123 124 125 126 127 128	GND ↓ 5V GND	GND ↓ 5V 5V GND	GND ↓ 5V 5V GND	GND GND 5V 5V GND ↓ 5V 5V 5V	IN ↓	 0 IT 0 IT 0 IT 0 IT	 OUT OUT OUT OUT	GND	 OUT OUT OUT OUT	GND ↓	GND ↓ 5V GND ↓	GND ↓ 5V GND ↓	GND ↓ 5V 5V GND ↓	OUT OUT	OUT OUT	5V ↓	E1 to E0 E1 to E0 E1 to GS E1 to GS E1 to Q0 E1 to Q1 E1 to Q0 E1 to Q1 E1 to Q2 E1 to Q2 E1 to Q0 E1 to Q1 E1 to Q2 E1 to Q0 E1 to Q1 E1 to Q2	11 ↓ 17	220 ↓ 340	16 ↓ 24	310 ↓ 480	11 ↓ 17	220 ↓ 340	ns
t_{PHL4} & t_{PLH4} t_{PHL5} & t_{PLH5}		129 130 131 132 133 134 135 136	 IN GND	 IN GND	 IN GND	GND ↓ IN	5V					IN GND	IN GND	IN GND	IN GND	OUT		5V	D0 to GS D1 to GS D2 to GS D3 to GS D4 to GS D5 to GS D6 to GS D7 to GS							
		137 138 139 140 141 142 143 144 145 146 147 148	 IN GND	 IN IN GND	 IN IN GND	GND ↓ IN IN IN		 OUT OUT OUT OUT	OUT OUT OUT OUT		OUT OUT OUT OUT		IN GND	IN GND	IN IN GND			5V	D1 to Q0 D2 to Q1 D3 to Q0 D3 to Q1 D4 to Q2 D5 to Q0 D5 to Q2 D6 to Q2 D6 to Q1 D7 to Q0 D7 to Q1 D7 to Q2	22	440	31	620	22	440	

See notes at end of device type.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 9 T_A = 25°C		Subgp 10 T_A = 125°C		Subgp 11 T_A = -55°C		
		Symbol	D4	D5	D6	D7	E1	Q2	Q1	V_{SS}	Q0	D0	D1	D2	D3	GS	E0	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
t_{THL} & t_{TLH}	3004 Fig 7	149 150 151	GND	GND	GND	5.0V	IN	OUT	OUT	GND	OUT	GND	GND	GND	GND			5V	Q0 Q1 Q2	10	200	14	280	10	200	ns
																				Subgp 12 T_A = 25°C						
																				Min	Max					
t_{PHL1} t_{PLH1} t_{PHL2} t_{PLH2} t_{PHL3}	3003 Fig 7	152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167	GND 10.0 GND	GND 10.0 10.0 GND	GND 10.0 10.0 GND	GND GND 10.0 10.0 GND 10.0 10.0 10.0	IN	OUT OUT OUT OUT	OUT OUT OUT OUT	GND	OUT OUT OUT OUT	GND 10.0 GND	GND 10.0 GND	GND 10.0 GND	GND 10.0 10.0 GND	OUT OUT	OUT OUT	10.0V	EI to EO EI to EO EI to GS EI to GS EI to Q0 EI to Q1 EI to Q0 EI to Q1 EI to Q2 EI to Q2 EI to Q0 EI to Q1 EI to Q2 EI to Q0 EI to Q1 EI to Q2	6 9	110 170					ns
t_{PLH3}		168 169 170 171 172 173 174 175 176 177 178 179	10.0 GND	10.0 10.0 GND	10.0 10.0 GND	GND 10.0 10.0 10.0		OUT OUT OUT OUT	OUT OUT OUT OUT		OUT OUT OUT OUT	10.0	10.0						EI to Q0 EI to Q1 EI to Q0 EI to Q1 EI to Q2 EI to Q2 EI to Q0 EI to Q1 EI to Q2 EI to Q0 EI to Q1 EI to Q2							
t_{PHL4}	3003 Fig 7	180 181 182 183 184 185 186 187	IN GND GND GND	IN GND GND GND	IN GND GND GND	GND	10.0					IN GND	GND IN GND	IN GND	IN GND	OUT			D0 to GS D1 to GS D2 to GS D3 to GS D4 to GS D5 to GS D6 to GS D7 to GS							
t_{PLH4}		188 189 190 191 192 193 194 195	GND IN GND	GND IN GND GND	GND IN GND	IN						IN GND	IN GND	GND IN GND	IN GND				D0 to GS D1 to GS D2 to GS D3 to GS D4 to GS D5 to GS D6 to GS D7 to GS							

See notes at end of device type.

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TABLE III. Group A inspection for device type 03 - Continued.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 12 T_A = 25°C						
		Symbol	D4	D5	D6	D7	E1	Q2	Q1	V_{SS}	Q0	D0	D1	D2	D3	GS	E0	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
t_{PHL5}	3003 Fig 7	196	GND	GND	GND	GND	10.0		OUT	GND	OUT	GND	IN	GND	GND			10.0V	D1 to Q0	11	220					ns
		197									OUT		GND	IN	GND				D2 to Q1							
		198							OUT					GND	IN				D3 to Q0							
		199													IN				D3 to Q1							
		200	IN					OUT							GND				D4 to Q2							
		201	GND	IN							OUT								D5 to Q0							
		202		IN				OUT											D5 to Q2							
		203		GND	IN			OUT											D6 to Q2							
		204			IN				OUT										D6 to Q1							
		205			GND	IN					OUT								D7 to Q0							
		206							OUT										D7 to Q1							
		207						OUT											D7 to Q2							
t_{PLH5}		208				GND			OUT		OUT		IN						D1 to Q0							
		209											GND						D2 to Q1							
		210							OUT		OUT			IN	IN				D3 to Q0							
		211												GND	GND				D3 to Q1							
		212	IN					OUT											D4 to Q2							
		213	GND	IN							OUT								D5 to Q0							
		214		IN				OUT											D5 to Q2							
		215		GND	IN			OUT											D6 to Q2							
		216			IN				OUT										D6 to Q1							
		217			GND	IN					OUT								D7 to Q0							
		218				IN			OUT										D7 to Q1							
		219				IN		OUT											D7 to Q2							
t_{THL} / t_{TLH}	3004 Fig 7	220				10.0	IN		OUT		OUT								Q0	5	100					
		221																	Q1							
		222						OUT											Q2							
		223							OUT		OUT								Q0							
		224																	Q1							
		225						OUT											Q2							

1/ Pins not designated may be High level logic, Low level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The following timing sequence and input/output conditions shall apply: These tests require only a summary of attributes data.

TIME SLOT	Inputs									Outputs				
	$\overline{E1}$	$\overline{D7}$	$\overline{D6}$	$\overline{D5}$	$\overline{D4}$	$\overline{D3}$	$\overline{D2}$	$\overline{D1}$	$\overline{D0}$	$\overline{GS}$	$\overline{Q2}$	$\overline{Q1}$	$\overline{Q0}$	$\overline{E0}$
1	0	X	X	X	X	X	X	X	X	0	0	0	0	0
2	1	0	0	0	0	0	0	0	0	0	0	0	0	1
3	1	1	X	X	X	X	X	X	X	1	1	1	1	0
4	1	0	1	X	X	X	X	X	X	1	1	1	0	0
5	1	0	0	1	X	X	X	X	X	1	1	0	1	0
6	1	0	0	0	1	X	X	X	X	1	1	0	0	0
7	1	0	0	0	0	1	X	X	X	1	0	1	1	0
8	1	0	0	0	0	0	1	X	X	1	0	1	0	0
9	1	0	0	0	0	0	0	1	X	1	0	0	1	0
10	1	0	0	0	0	0	0	0	1	1	0	0	0	0

4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for individual input or measure all inputs together.

5/ See 4.4.1c.

6/ The Truth Table Tests shall be performed in sequence.

7/ The Truth Table Tests shall be performed at V_{IH} and $V_{DD} \geq$ 5 Vdc and $\leq$ 18 Vdc. L = 0.50V maximum and [illegible] V_{SS} -0.50V minimum).

Test	V_{DD}	Input/output conditions: Input levels		Output levels	
		1	0	1	0
	Volts			Min	Max
V_{OH1} / V_{OL1}	15.0	15.00	GND	14.95V	0.05V
V_{IH1} / V_{IL1}	5.0	3.5V	1.5V	4.5V	0.5V
V_{IH2} / V_{IL2}	10.0	7.0V	3.0V	9.0V	1.0V
V_{IH3} / V_{IL3}	15.0	11.0V	4.0V	13.5V	1.5V

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TABLE III. Group A inspection for device type 04.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test Limits Subgp 1 T_A = 25°C Min	Max	Subgp 2 T_A = 125°C Min	Max	Subgp 3 T_A = -55°C Min	Max	Units
		Symbol	$\overline{TE}$	1A	1B	1Q0	1Q1	1Q2	1Q3	V_{SS}	2Q3	2Q2	2Q1	2Q0	2B	2A	$\overline{2E}$	V_{DD}								
		Test No.																								
V_{IC} (pos)		1 2 3 4 5 6	1mA	1mA	1mA										1mA	1mA	1mA	GND ↓	$\overline{TE}$ 1A 1B 2B 2A $\overline{2E}$		1.5 ↓					Vdc ↓
V_{IC} (neg)		7 8 9 10 11 12	-1mA	-1mA	-1mA					GND ↓					-1mA	-1mA	-1mA		$\overline{TE}$ 1A 1B 2B 2A $\overline{2E}$		-6.0 ↓					↓
I_{SS} 2/		13 14 15 16 17	GND ↓ 18V	GND GND 18V ↓	GND 18V GND 18V 18V										GND 18V GND 18V 18V	GND GND 18V ↓	GND ↓ 18V	18V ↓	V_{SS} ↓		-.25 ↓		-2.5 ↓			µA ↓
V_{OL1} ↓	3007 ↓	18 19 20 21 22 23 24 25 26 27	GND ↓ 15V GND	GND 15V GND 15V 15V GND ↓	GND GND 15V 15V 15V GND ↓										GND ↓ 15V 15V 15V	GND ↓ 15V GND 15V 15V	GND ↓ 15V	15V ↓	1Q3 1Q2 1Q1 1Q0 1Q3 2Q3 2Q2 2Q1 2Q0 2Q3		0.05 ↓		0.05 ↓		0.05 ↓	Vdc ↓
V_{OH1} ↓	3006 ↓	28 29 30 31 32 33 34 35		15V GND 15V GND ↓	15V 15V GND ↓										GND ↓ 15V 15V	GND ↓ 15V GND 15V	GND ↓		1Q0 1Q1 1Q2 1Q3 2Q0 2Q1 2Q2 2Q3	14.95 ↓		14.95 ↓		14.95 ↓		↓
V_{IL1} V_{IL2} V_{IL3} V_{IH1} V_{IH2} V_{IH3} ↓		36 37 38 39 40 41	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓		3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	All outputs ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	3/ ↓	
I_{OL1} ↓		42 43 44 45 46 47 48 49	5V ↓	GND ↓	GND ↓	0.4V	0.4V	0.4V	0.4V	↓	0.4V	0.4V	0.4V	0.4V	GND ↓	GND ↓	5V ↓	5V ↓	1Q0 1Q1 1Q2 1Q3 2Q3 2Q2 2Q1 2Q0	0.51 ↓		0.36 ↓		0.64 ↓		nA ↓

See notes at end of device type.

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Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp3 T_A = -55°C		
		Symbol	$\overline{TE}$	1A	1B	1Q0	1Q1	1Q2	1Q3	V_{SS}	2Q3	2Q2	2Q1	2Q0	2B	2A	$\overline{2E}$	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
I_{OL2}		50 51 52 53 54 55 56 57	15V	GND	GND	1.5V	1.5V	1.5V	1.5V	GND	1.5V	1.5V	1.5V	1.5V	GND	GND	15V	15V	1Q0 1Q1 1Q2 1Q3 2Q3 2Q2 2Q1 2Q0	3.4		2.4		4.2		mA
I_{OH1}		58 59 60 61 62 63 64 65	GND	GND 5V GND 5V	GND GND 5V	4.6V	4.6V	4.6V	4.6V		4.6V	4.6V	4.6V	4.6V	5V GND	5V GND	GND	5V	1Q0 1Q1 1Q2 1Q3 2Q3 2Q2 2Q1 2Q0	-0.51		-0.36		-0.64		
I_{OH2}		66 67 68 69 70 71 72 73		GND 15V GND 15V	GND GND 15V	13.5V	13.5V	13.5V	13.5V		13.5V	13.5V	13.5V	13.5V	15V GND	15V GND		15V	1Q0 1Q1 1Q2 1Q3 2Q3 2Q2 2Q1 2Q0	-3.4		-2.4		-4.2		
I_{IH1} 4/	3010	74	18V	18V	18V										18V	18V	18V	18V	All inputs together		6					nA
I_{IH2}		75 76 77 78 79 80	18V GND	GND 18V GND	GND GND 18V GND										GND 18V GND GND	GND 18V GND	GND 18V		$\overline{TE}$ 1A 1B 2B 2A $\overline{2E}$		1		45			
I_{IL1} 4/	3009	81	GND	GND	GND										GND	GND	GND		All inputs together		-6					
I_{IL2}		82 83 84 85 86 87	GND 18V	18V GND 18V 18V	18V 18V GND 18V										18V GND 18V 18V	18V GND 18V	18V GND		$\overline{TE}$ 1A 1B 2B 2A $\overline{2E}$		-1		-45			
																				Subgp 4 T_A = 25°C						
																				Min	Max					
C_1	3012	88 89 90 91 92 93	5/	5/	5/					GND					5/	5/	5/	GND	$\overline{TE}$ 1A 1B 2B 2A $\overline{2E}$		7.5					pF

See notes at end of device type.

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/															Measured terminal	Test limits						Units
		E,F,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 7 T_A = 25°C		Subgp 8 T_A = 125°C		T_A = -55°C		
		Symbol	$\overline{TE}$	1A	1B	1Q0	1Q1	1Q2	1Q3	V_{SS}	2Q3	2Q2	2Q1	2Q0	2B	2A	$\overline{2E}$	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
TRUTH TABLE TEST	3014	94 95 96 97 98 99 100 101 102 103	GND ↓ 5V	GND 5V GND 5V 5V	GND GND 5V 5V 5V	H L ↓	L H L ↓	L L H L L	L ↓ H L	GND ↓	L L L H L	L L H L L	L H L ↓	H L ↓	GND GND 5V ↓	GND 5V GND 5V 5V	GND ↓ 5V	5V ↓	All outputs ↓	See 6 7/						
																				Subgp 9 T_A = 25°C		Subgp 10 T_A = 125°C		Subgp 11 T_A = -55°C		
																				Min	Max	Min	Max	Min	Max	
t_{PHL1} & t_{PLH1}	3003 Fig 7	104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119	GND ↓	GND GND IN IN 5V 5V IN IN GND ↓	IN IN GND GND IN IN 5V 5V GND ↓	OUT OUT	OUT OUT	OUT OUT	OUT OUT		OUT OUT	OUT OUT	OUT OUT	OUT OUT	GND ↓ IN IN GND GND IN IN 5V 5V	GND ↓ IN IN 5V 5V IN IN	GND ↓		1B to 1Q0 1B to 1Q1 1A to 1Q2 1A to 1Q0 1B to 1Q1 1B to 1Q3 1A to 1Q2 1A to 1Q3 2B to 2Q0 2B to 2Q1 2A to 2Q2 2A to 2Q0 2B to 2Q1 2B to 2Q3 2A to 2Q2 2A to 2Q3	22	440	31	620	22	440	ns
t_{PHL2} & t_{PLH2}		120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135	IN ↓ GND ↓ IN ↓ GND ↓	GND 5V GND 5V GND ↓ 5V GND 5V GND ↓	GND GND 5V 5V GND ↓ 5V 5V GND ↓	OUT OUT	OUT OUT	OUT OUT	OUT OUT		OUT OUT	OUT OUT	OUT OUT	OUT OUT	GND ↓ 5V 5V GND ↓ 5V 5V	GND ↓ 5V GND 5V GND ↓ 5V GND 5V	GND ↓ IN ↓ GND ↓ IN ↓		1E to 1Q0 1E to 1Q1 1E to 1Q2 1E to 1Q3 2E to 2Q0 2E to 2Q1 2E to 2Q2 2E to 2Q3 1E to 1Q0 1E to 1Q1 1E to 1Q2 1E to 1Q3 2E to 2Q0 2E to 2Q1 2E to 2Q2 2E to 2Q3	20	400	28	560	20	400	
t_{THL} & t_{TLH}	3004 Fig 7	136 137 138 139 140 141 142 143	IN ↓ GND ↓	GND 5V GND 5V GND ↓	GND GND 5V 5V GND ↓	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	GND ↓ 5V 5V	GND ↓ 5V GND 5V	GND ↓ IN ↓		1Q0 1Q1 1Q2 1Q3 2Q0 2Q1 2Q2 2Q3	10	200	14	280	10	200	

See notes at end of device type.

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TABLE III. Group A inspection for device type 04 - Continued.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgp 12 T_A = 25°C Min	Max	Min	Max	Min	Max	Units
		Symbol	$\overline{TE}$	1A	1B	1Q0	1Q1	1Q2	1Q3	V_{SS}	2Q3	2Q2	2Q1	2Q0	2B	2A	$\overline{2E}$	V_{DD}								
		Test No.																								
t_{PHL1} & t_{PLH1}	3003 Fig 7	144	GND	GND	IN	OUT				GND					GND	GND	GND	10.0V	1B to 1Q0	10	190					ns
		145		GND	IN		OUT												1B to 1Q1							
		146		IN	GND	OUT		OUT											1A to 1Q2							
		147		IN	GND		OUT												1A to 1Q0							
		148		10.0	IN				OUT										1B to 1Q1							
		149		10.0	IN			OUT											1B to 1Q3							
		150		IN	10.0				OUT										1A to 1Q2							
		151		IN	10.0														1A to 1Q3							
		152		GND	GND														2B to 2Q0							
		153																	2B to 2Q1							
		154																	2A to 2Q2							
		155																	2A to 2Q0							
		156																	2B to 2Q1							
		157																	2B to 2Q3							
		158																	2A to 2Q2							
		159																	2A to 2Q3							
t_{PHL2} & t_{PLH2}		160	IN	GND	GND	OUT									GND	GND			$\overline{1E}$ to 1Q0	9	170					
		161		10.0	GND		OUT												$\overline{1E}$ to 1Q1							
		162		GND	10.0			OUT											$\overline{1E}$ to 1Q2							
		163		10.0	10.0				OUT										$\overline{1E}$ to 1Q3							
		164	GND	GND	GND														$\overline{2E}$ to 2Q0							
		165																	$\overline{2E}$ to 2Q1							
		166																	$\overline{2E}$ to 2Q2							
		167																	$\overline{2E}$ to 2Q3							
t_{THL} & t_{TLH}	3004 Fig 7	168	IN	10.0		OUT									GND	GND	GND		1Q0	5	100					
		169		GND	10.0		OUT												1Q1							
		170		10.0	10.0			OUT											1Q2							
		171		GND	GND				OUT										1Q3							
		172																	2Q0							
		173																	2Q1							
		174																	2Q2							
		175	GND								OUT	OUT	OUT	OUT	10.0	10.0	IN		2Q3							

1/ Pins not designated may be High level logic, Low level logic or Open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The following timing sequence and input/output conditions shall apply: These tests require only a summary of attributes data.

Inputs Enable	select		Outputs Device type 04			
$\overline{E}$	B	A	Q3	Q2	Q1	Q0
0	0	0	0	0	0	1
0	0	1	0	0	1	0
0	1	0	0	1	0	0
0	1	1	1	0	0	0
1	X	X	0	0	0	0

4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for individual input or measure all inputs together.

5/ See 4.4.1c.

6/ The Truth Table Tests shall be performed in sequence.

7/ The Truth Table Tests shall be performed at V_{IH} and $V_{DD} \leq 5$ Vdc and ≥ 18 Vdc. L = V_{SS} +0.50 V maximum and H = V_{DD} -0.50 V minimum.

Input/output conditions

Test	V_{DD}	Input levels 1	0	Output levels 1 Min	0 Max
	Volts				
V_{IH1} V_{IL1}	5.0	3.5V	1.5V	4.5V	0.5V
V_{IH2} V_{IL2}	10.0	7.0V	3.0V	9.0V	1.0V
V_{IH3} V_{IL3}	15.0	11.0V	4.0V	13.5V	1.5V

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TABLE III. Group A inspection for device type 05.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/																Measured terminal	Test limits						Units
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	$\overline{TE}$	1A	1B	$\overline{TQ0}$	$\overline{TQ1}$	$\overline{TQ2}$	$\overline{TQ3}$	V_{SS}	$\overline{2Q3}$	$\overline{2Q2}$	$\overline{2Q1}$	$\overline{2Q0}$	2B	2A	$\overline{2E}$	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
V_{IC} (pos)		1 2 3 4 5 6	1mA	1mA	1mA										1mA	1mA	1mA	GND	$\overline{TE}$ 1A 1B 2B 2A $\overline{2E}$		1.5					Vdc
V_{IC} (neg)		7 8 9 10 11 12	-1mA	-1mA	-1mA					GND					-1mA	-1mA	-1mA		$\overline{TE}$ 1A 1B 2B 2A $\overline{2E}$		-6.0					Vdc
I_{SS} 2/		13 14 15 16 17	GND 18V	GND GND 18V	GND 18V 18V										GND 18V GND 18V 18V	GND GND 18V	GND 18V	18V	V_{SS}		-.25		-2.5			µA
V_{OL1}	3007	18 19 20 21 22 23 24 25	GND	GND 15V GND 15V GND	GND GND 15V 15V GND										GND 15V 15V	GND 15V GND 15V	GND	15V	$\overline{TQ0}$ $\overline{TQ1}$ $\overline{TQ2}$ $\overline{TQ3}$ $\overline{2Q0}$ $\overline{2Q1}$ $\overline{2Q2}$ $\overline{2Q3}$		0.05		0.05		0.05	Vdc
V_{OH1}	3006	26 27 28 29 30 31 32 33 34 35	15V GND	15V GND 15V 15V GND	15V 15V 15V 15V GND										GND 15V	GND 15V GND 15V 15V	15V	15V	$\overline{TQ3}$ $\overline{TQ2}$ $\overline{TQ1}$ $\overline{TQ0}$ $\overline{2Q0}$ $\overline{2Q3}$ $\overline{2Q2}$ $\overline{2Q1}$ $\overline{2Q0}$ $\overline{2Q0}$	14.95		14.95		14.95		
V_{IL1} V_{IL2} V_{IL3} V_{IH1} V_{IH2} V_{IH3}		36 37 38 39 40 41	3/	3/	3/	3/	3/	3/	3/		3/	3/	3/	3/	3/	3/	3/	3/	All outputs	3/	3/	3/	3/	3/	3/	
I_{OL1}		42 43 44 45 46 47 48 49	GND	GND 5V GND 5V GND	GND GND 5V 5V GND	0.4V	0.4V	0.4V	0.4V		0.4V	0.4V	0.4V	0.4V	GND 5V 5V GND GND	GND 5V GND 5V GND	GND	5V	$\overline{TQ0}$ $\overline{TQ1}$ $\overline{TQ2}$ $\overline{TQ3}$ $\overline{2Q3}$ $\overline{2Q2}$ $\overline{2Q1}$ $\overline{2Q0}$	0.51		0.36		0.64		mA
I_{OL2}		50 51 52 53 54 55 56 57		15V GND 15V GND	15V 15V 15V GND	1.5V	1.5V	1.5V	1.5V		1.5V	1.5V	1.5V	1.5V	GND 15V 15V GND GND	GND 15V GND 15V GND		15V	$\overline{TQ0}$ $\overline{TQ1}$ $\overline{TQ2}$ $\overline{TQ3}$ $\overline{2Q3}$ $\overline{2Q2}$ $\overline{2Q1}$ $\overline{2Q0}$	3.4		2.4		4.2		

See notes at end of device type.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Units
		E,F,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 1 T_A = 25°C		Subgp 2 T_A = 125°C		Subgp 3 T_A = -55°C		
		Symbol	$\overline{TE}$	1A	1B	$\overline{TQ0}$	$\overline{TQ1}$	$\overline{TQ2}$	$\overline{TQ3}$	V_{SS}	$\overline{2Q3}$	$\overline{2Q2}$	$\overline{2Q1}$	$\overline{2Q0}$	2B	2A	$\overline{2E}$	V_{DD}		Min	Max	Min	Max	Min	Max	
		Test No.																								
I_{OH1}		58 59 60 61 62 63 64 65	5V	GND	GND	4.6V	4.6V	4.6V	4.6V	GND	4.6V	4.6V	4.6V	4.6V	GND	GND	5V	5V	$\overline{Q0}$ $\overline{Q1}$ $\overline{Q2}$ $\overline{Q3}$ $\overline{2Q3}$ $\overline{2Q2}$ $\overline{2Q1}$ $\overline{2Q0}$	-0.51		-0.36		-0.64		mA
I_{OH2}		66 67 68 69 70 71 72 73	15V			13.5V	13.5V	13.5V	13.5V		13.5V	13.5V	13.5V	13.5V			15V	15V		-3.4		-2.4		-4.2		
I_{IH1} 4/	3010	74	18V	18V	18V					GND					18V	18V	18V	18V	All inputs together		6					nA
I_{IH2}	3010	75 76 77 78 79 80	18V GND	GND 18V GND	GND GND 18V GND										GND GND GND 18V GND GND	GND 18V GND	GND 18V	18V	1E 1A 1B 2B 2A 2E		1		45			
I_{IL1} 4/	3009	81	GND	GND	GND					GND					GND	GND	GND		All inputs together		-6					
I_{IL2}	3009	82 83 84 85 86 87	GND 18V	18V GND 18V	18V 18V GND 18V					GND					18V 18V 18V GND 18V 18V	18V GND 18V	18V GND		1E 1A 1B 2B 2A 2E		-1		-45			
																				Subgp 4 T_A = 25°C						
																				Min	Max					
C_I	3012	88 89 90 91 92 93	5/	5/	5/					GND					5/	5/	5/	GND	1E 1A 1B 2B 2A 2E		7.5					pF

See notes at end of device type.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Case E,F,Z	Terminal conditions 1/ 1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Test limits Subgp 7 T_A = 25°C Min	Max	Subgp 8 T_A = 125°C Min	Max	T_A = -55°C Min	Max	Units
		Symbol	$\overline{TE}$	1A	1B	$\overline{1Q0}$	$\overline{1Q1}$	$\overline{1Q2}$	$\overline{1Q3}$	V_{SS}	$\overline{2Q3}$	$\overline{2Q2}$	$\overline{2Q1}$	$\overline{2Q0}$	2B	2A	$\overline{2E}$	V_{DD}								
		Test No.																								
TRUTH TABLE TEST	3014	94	GND	GND	GND	L	H	H	H	GND	H	H	H	L	GND	GND	GND	5V	All outputs	See 6/ 7/						
		95	↓	5V	GND	H	L	H	H	↓	H	H	H	H	GND	5V	↓	↓	↓								
		96		GND	5V	H	H	L	H		H	L	H	H	5V	GND											
		97		5V	5V	H	H	H	L		L	H	H	H	5V	5V											
		98	5V	5V	5V	H	H	H	H		H	H	H	H	5V	5V	5V										
																				Subgp 9 T_A = 25°C Min	Max	Subgp 10 T_A = 125°C Min	Max	Subgp 11 T_A = -55°C Min	Max	
t_{PHL1} & t_{PLH1}	3003 Fig 7	99	GND	GND	IN		OUT			GND					GND	GND	GND	5V	1B to $\overline{1Q0}$	22	440	31	620	22	440	ns
		100		5V	IN	OUT													1B to $\overline{1Q2}$							
		101		IN	GND		OUT	OUT											1A to $\overline{1Q1}$							
		102		IN	GND	OUT													1A to $\overline{1Q0}$							
		103		5V	IN		OUT												1B to $\overline{1Q1}$							
		104		5V	IN			OUT	OUT										1B to $\overline{1Q3}$							
		105		IN	5V														1A to $\overline{1Q2}$							
		106		IN	5V			OUT	OUT										1A to $\overline{1Q3}$							
		107		IN	GND								OUT		IN				2B to $\overline{2Q0}$							
		108		GND	↓									OUT	IN				2B to $\overline{2Q2}$							
		109										OUT			GND	IN			2A to $\overline{2Q1}$							
		110												OUT	GND	IN			2A to $\overline{2Q0}$							
		111											OUT		IN	5V			2B to $\overline{2Q1}$							
		112													IN	5V			2B to $\overline{2Q3}$							
		113									OUT	OUT			5V	IN			2A to $\overline{2Q2}$							
		114									OUT				5V	IN			2A to $\overline{2Q3}$							
t_{PHL2} & t_{PLH2}		115	IN	GND	GND	OUT									GND	GND			$\overline{TE}$ to $\overline{1Q0}$	20	400	28	560	20	400	
		116		5V	GND		OUT												$\overline{TE}$ to $\overline{1Q1}$							
		117		GND	5V			OUT											$\overline{TE}$ to $\overline{1Q2}$							
		118		5V	5V				OUT										$\overline{TE}$ to $\overline{1Q3}$							
		119	GND	GND	GND														2E to $\overline{2Q0}$							
		120											OUT	OUT		5V	IN		2E to $\overline{2Q1}$							
		121										OUT				GND			2E to $\overline{2Q2}$							
		122									OUT				5V	5V			2E to $\overline{2Q3}$							
t_{THL} & t_{TLH}	3004 Fig 7	123	IN	5V	5V	OUT									GND	GND	GND		$\overline{1Q0}$	10	200	14	280	10	200	
		124		GND	5V		OUT												$\overline{1Q1}$							
		125		5V	5V			OUT											$\overline{1Q2}$							
		126		GND	GND				OUT										$\overline{1Q3}$							
		127	GND																$\overline{2Q0}$							
		128											OUT	OUT		5V	IN		$\overline{2Q1}$							
		129										OUT				GND			$\overline{2Q2}$							
		130									OUT				5V	5V			$\overline{2Q3}$							

See notes at end of device type.

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TABLE III. Group A inspection for device type 05 - Continued.

Symbol	MIL-STD-883 method	Case	Terminal conditions 1/																Measured terminal	Test limits						Units
		E,F,Z	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16		Subgp 12 T_A = 25°C					
		Symbol	$\overline{1E}$	1A	1B	$\overline{1Q0}$	$\overline{1Q1}$	$\overline{1Q2}$	$\overline{1Q3}$	V_{SS}	$\overline{2Q3}$	$\overline{2Q2}$	$\overline{2Q1}$	$\overline{2Q0}$	2B	2A	$\overline{2E}$	V_{DD}								
		Test No.																		Min	Max	Min	Max	Min	Max	
t_{PHL1} & t_{PLH1}	3003 Fig 7	131	GND	GND	IN	OUT				GND					GND	GND	GND	10.0	1B to $\overline{1Q0}$	10	190					ns
		132		GND	IN		OUT												1B to $\overline{1Q1}$							
		133		IN	GND			OUT											1A to $\overline{1Q2}$							
		134		IN	GND	OUT													1A to $\overline{1Q0}$							
		135		10.0	IN		OUT												1B to $\overline{1Q1}$							
		136		10.0	IN				OUT										1B to $\overline{1Q3}$							
		137		IN	10.0			OUT											1A to $\overline{1Q2}$							
		138		IN	10.0				OUT										1A to $\overline{1Q3}$							
		139		GND	GND									OUT	IN				2B to $\overline{2Q0}$							
		140											OUT		IN				2B to $\overline{2Q1}$							
		141										OUT			GND	IN			2A to $\overline{2Q2}$							
		142												OUT	GND	IN			2A to $\overline{2Q0}$							
		143									OUT		OUT		IN	10.0			2B to $\overline{2Q1}$							
		144													IN	10.0			2B to $\overline{2Q3}$							
		145										OUT			10.0	IN			2A to $\overline{2Q2}$							
		146									OUT				10.0	IN			2A to $\overline{2Q3}$							
t_{PHL2} & t_{PLH2}		147	IN	GND	GND	OUT									GND	GND			$\overline{1E}$ to $\overline{1Q0}$	9	170					
		148		10.0	GND		OUT												$\overline{1E}$ to $\overline{1Q1}$							
		149		GND	10.0			OUT											$\overline{1E}$ to $\overline{1Q2}$							
		150		10.0	10.0				OUT										$\overline{1E}$ to $\overline{1Q3}$							
		151	GND	GND	GND												IN		$\overline{2E}$ to $\overline{2Q0}$							
		152											OUT			10.0			$\overline{2E}$ to $\overline{2Q1}$							
		153										OUT		OUT	10.0	GND			$\overline{2E}$ to $\overline{2Q2}$							
		154									OUT				10.0	10.0			$\overline{2E}$ to $\overline{2Q3}$							
t_{THL} & t_{TLH}	3004 Fig 7	155	IN	10.0		OUT									GND	GND	GND		$\overline{1Q0}$	5	100					
		156		GND	10.0		OUT												$\overline{1Q1}$							
		157		10.0	10.0			OUT											$\overline{1Q2}$							
		158		GND	10.0				OUT										$\overline{1Q3}$							
		159	GND	GND	GND												IN		$\overline{2Q0}$							
		160											OUT			10.0			$\overline{2Q1}$							
		161										OUT		OUT	10.0	GND			$\overline{2Q2}$							
		162									OUT				10.0	10.0			$\overline{2Q3}$							

1/ Pins not designated may be High level logic, low level logic or open. Exceptions are as follows: $V_{IC(pos)}$ tests, the V_{SS} terminal shall be open; $V_{IC(neg)}$ tests, the V_{DD} terminal shall be open; I_{SS} tests, the output terminals shall be open.

2/ The I_{SS} measurements shall be performed in sequence.

3/ The following timing sequence and input/output conditions shall apply: These tests require only a summary of attributes data.

Inputs Enable	select		Outputs Device type 05			
$\overline{E}$	B	A	$\overline{Q3}$	$\overline{Q2}$	$\overline{Q1}$	$\overline{Q0}$
0	0	0	1	1	1	0
0	0	1	1	1	0	1
0	1	0	1	0	1	1
0	1	1	0	1	1	1
1	X	X	1	1	1	1

4/ The device manufacturer may, at his option, measure I_{IL} and I_{IH} at 25°C for individual input or measure all inputs together.

5/ See 4.4.1c.

6/ The Truth Table Tests shall be performed in sequence.

7/ The Truth Table Tests shall be performed at V_{IH} and $V_{DD} \leq 5$ Vdc and ≥ 18 Vdc. L = V_{SS} +0.50 V maximum and H = V_{DD} -0.50 V minimum.

Input/output conditions

Test	V_{DD}	Input levels		Output levels	
		1	0	1	0
	Volts			Min	Max
V_{IH1} V_{IL1}	5.0	3.5V	1.5V	4.5V	0.5V
V_{IH2} V_{IL2}	10.0	7.0V	3.0V	9.0V	1.0V
V_{IH3} V_{IL3}	15.0	11.0V	4.0V	13.5V	1.5V

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d. For class S devices, post dynamic burn-in, or class B devices, post static burn-in, electrical parameter measurements may, at the manufacturer's option, be performed separately or included in the final electrical parameter requirements.

4.2.1 Percent defective allowable (PDA).

a. The class S devices PDA shall be 5 percent for static burn-in and 5 percent for dynamic burn-in, based on the exact number of devices submitted to each separate burn-in.

b. Static burn-in I and II failures shall be cumulative for determining PDA.

c. The class B devices PDA shall be in accordance with MIL-M-38510 for static burn-in. Dynamic burn-in is not required.

d. Those devices whose measured characteristics after burn-in exceed the specified delta (Δ) limits or electrical parameter limits specified in table III, subgroup 1, are defective and shall be removed from the lot. The verified failures divided by the total number of devices in lot initially submitted to burn-in shall be used to determine the percent defective for the lot and the lot shall be accepted or rejected based on the specified PDA.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D and E inspections (see 4.4.1 through 4.4.5).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510 and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, D, and E inspections (see 4.4.1 through 4.4.5).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be performed in accordance with table II herein.

b. Subgroups 5 and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

c. Subgroup 4 (C_i measurement) shall be measured only for initial qualification and after process or design changes which may affect input capacitance. Capacitance shall be measured between the designated terminal and V_{SS} at a frequency of 1 MHz.

d. Subgroup 12 shall be added to the group A inspection requirements for class S devices using an LTPD of 15 and consist of the procedures, test conditions, and limits specified in table III.

e. At the manufacturer's option, test tapes may be programmed simultaneously for each identical section provided that each output is measured and each specified input combination is tested.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. Class S steady state life (accelerated) test circuits shall be submitted to the qualifying activity for approval. When the alternate steady state life test is used, the circuit on figure 6, or equivalent, shall be used.

b. A special subgroup shall be added using an LTPD of 15 for classes S and B, and measured only for initial qualification and after process or design changes. This subgroup shall consist of a high voltage test of the input protection circuits, V_{ZAP} (see 4.5.3).

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c. End-point electrical parameters shall be as specified in table II herein and shall consist only of those subgroups specified in table IIa of test method 5005 of MIL-STD-883, and table II herein also. Delta limits shall apply only to subgroup 5 of group B inspections and shall consist of tests specified in table IV herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection and shall consist of tests specified in table IV herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D and as specified in 4.5.2 and as shown on figure 6, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration, 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. Subgroups 3 and 4 shall be added to the group C inspection requirements for class B devices, and shall consist of the tests, conditions, and limits as specified for subgroups 10 and 11 of group A.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.4.5 Group E inspection. Group E inspection is required only for device types intended to be marked as radiation hardened (see 3.6.1). When group E testing is performed it shall be in accordance with table V of method 5005 of MIL-STD-883 and 4.5.5 herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit V_{SS} terminal, unless otherwise specified. Currents given are conventional current and positive when flowing into the referenced terminal.

4.5.2 Burn-in and life test cool down procedures. When these tests are completed and prior to removal of bias voltages, the devices under test (DUT) shall be cooled to a temperature of 25° ±3°C; then, electrical parameter end-point measurements shall be performed.

TABLE IV. Delta limits @25°C.

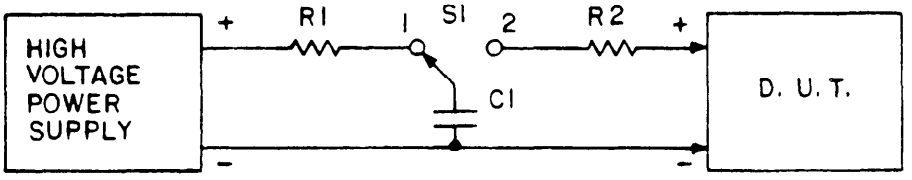
Parameter 1/	Device types
I_{SS}	±10 nA
I_{OL1}	15%
I_{OH1}	15%

1/ Each of the above parameters shall be recorded before and after the required burn-in and life tests to determine delta's (Δ).

4.5.3 High voltage (V_{ZAP}) test or input protection circuits. All input terminals (up to a maximum of 4) of the DUT shall be subjected to a voltage pulse from a 100 pF source charged to 400 V. This destructive test shall be conducted as follows using the test circuit on figure 8.

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a. Measure I_{IL} and I_{IH} at the inputs selected, as stated above, at 25°C. The test limit for each input tested shall be ±10 nA at the specified V_{DD}. Measure I_{SS} on the DUT at 25°C. The test limit for this measurement shall be increased a maximum of 20 percent of the specified I_{SS} table III limit at the specified V_{DD}.



V_{ZAP} = 400 V minimum charge on C1
1 MΩ ≤ R1 ≤ 50 MΩ
R2 = 1.5 kΩ
C1 = 100 pF
S1 = Hg-wetted "bounceless" relay.

FIGURE 8. High voltage (V_{ZAP}) test circuit.

b. V_{ZAP} is applied to DUT in the following modes (see table V) by changing C1 to V_{ZAP} with S1 in position 1 and then switching to position 2.

TABLE V. Modes for high voltage test.

Mode	+ Terminal	- Terminal
1	V_{DD}	Input
2	Input	V_{SS}
3	Input	Associated Output

c. Within 24 hours repeat the I_{SS}, I_{IL}, and I_{IH} measurements on the same terminals as performed above. If a DUT exhibits leakage currents in excess of the specified limits after the V_{ZAP} test, it shall be classified as a failure.

4.5.4 Quiescent supply current (I_{SS} test). When performing quiescent supply current measurements (I_{SS}), the meter shall be placed so that all currents flow through the meter.

4.5.5 Radiation hardness assurance (RHA) testing. The RHA testing shall be performed in accordance with test procedures and sampling specified in table V of method 5005 of MIL-STD-883 and herein:

a. Before irradiation, selected samples shall be assembled in qualified packages and pass the governing electrical parameters (group A subgroup 1 at 25°C) and also be subjected to the threshold-voltage test in table VIII in order to calculate the delta threshold (ΔV_T) after irradiation.

b. The devices shall be subjected to a total radiation dose as specified in MIL-M-38510 for the radiation hardness assurance (RHA) level being tested, and meet the end point electrical parameters as defined in table VI at 25°C, after exposure. The start and completion of the end point electrical parameter measurements shall not exceed 2 hours following irradiation.

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c. Threshold-voltage test circuit conditions shall be as specified in table VIII and figure 9. In situ and remote testing, the tests shall be performed with the devices biased in accordance with table VII and bias may be interrupted for up to 1 minute to remove devices to the remote bias fixture.

d. After irradiation, the devices shall pass the truth table test as specified in subgroup 7 in table III or if subgroup 7 is not required, then an equivalent truth table test shall be performed.

TABLE VI. Radiation hardening end-point electrical parameters at 25°C.

Parameter	All device types	V_{DD}
V_{TN}	0.3 V min	10 V
V_{TP}	2.8 V max	10 V
ΔV_T	1.4 V	10 V
I_{SS}	100 x max limit	18 V
t_{PLH}	1.35 x max limit	5 V
t_{PHL}	1.35 x max limit	5 V

TABLE VII. Bias during exposure to radiation.

Device type	Pin connections		
	V_{DD} = 10 V dc (through a 30 k to 60 k resistor)	V_{SS} = GND	V_{DD} = 10 V dc
01	1,2,3,21,22,23	12	14, 24
02	1,2,3,21,22,23	12	14, 24
03	1,2,3,4,5,10,11,12,13	8	16
04	1,2,3,13,14,15	8	16
05	1,2,3,13,14,15	8	16

Pins not designated are open or connected to 10 V dc through a 30- to 60 kilohm resistor.

4.6 Data reporting. When specified in the purchase order or contract, a copy of the following data, as applicable, shall be supplied.

a. Attributes data for all screening tests (see 4.2 and variables data for all static burn-in, dynamic burn-in, and steady state life tests (se 3.6).

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.6).

e. Final electrical parmeters data (see 4.2c).

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logisitic support of existing equipment.

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4.4.4 of method 5005 of MIL-STD-883, end-point electrical parameters shall be as

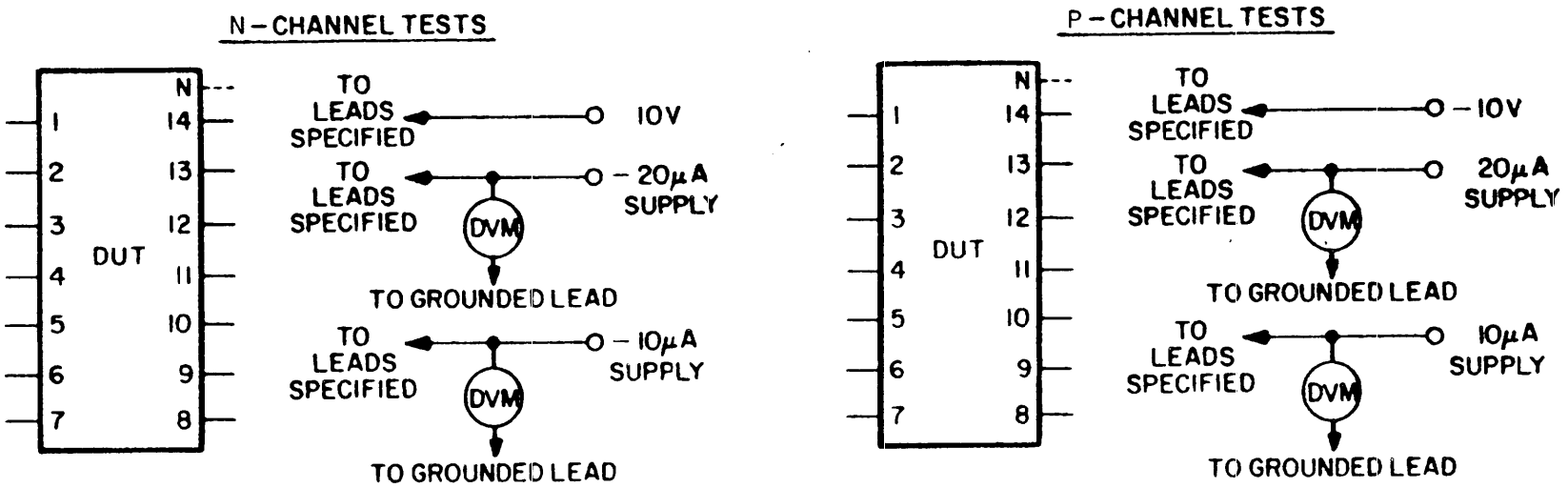


FIGURE 7. Threshold-voltage circuit.

TABLE VIII. Threshold - voltage test circuit conditions.

Device type	GND	10V	V_{TN} measured at		GND	-10V	V_{TP} measured at	
			-20 μA supply	-10 μA supply			20 μA supply	10 μA supply
01	2	24	3,12,21-23		1	2,3,12,21-23	24	
02	1	24	2,3,12,21-23		1	2,3,12,21-23	24	
03	1	2-5,10-13, 16		8	1	2-5,8,10,11, 12,13		16
04	1	16		2,3,8, 13-15	1	2,3,8, 13-15		16
05	1	16		2,3,8, 13-15	1	2,3,8, 13-15		16

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6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

i. Requirements for total dose radiation testing (see 3.6.1 and 4.5.5), if applicable.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

C_i	Input terminal-to-V_{SS} capacitance.
GND	Ground zero voltage potential.
I_{SS}	Quiescent supply current.
T_A	Free air temperature.
V_{DD}	Positive supply voltage.
V_{SS}	Negative supply voltage.
V_{ZAP}	Input test voltage.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class S for National Aeronautics and Space Administration or class B for Department of Defense (see 1.2.2), lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type	DESC drawing
01	4514B	7703501
02	4515B	7703201
03	4532B	
04	4555B	7704701
05	4556B	7704801

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6.6 Handling. MOS devices must be handled with certain precautions to avoid damage due to accumulation of static charge. Input protective devices have been designed in the chip to minimize the effect of this static build up. However, the following handling practices are recommended:

a. Devices should be handled on benches with conductive and grounded surface.

b. Ground test equipment and tools.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - MI
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM, AR
Navy - AS, CG, OC, MC, SH

Agent:
DLA - ES

Preparing activity:
NASA - NA

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-38510/173B	MICORCIRCUITS, DIGITAL, CMOS, DECODERS, ENCODER, AND DECODERS/DEMULTIPLEXERS MONOLITHIC SILICON, POSITIVE LOGIC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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